



Cyber risk and the U.S. financial system: A pre-mortem analysis[☆]

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ABSTRACT

We model how a cyber attack may be amplified through the U.S. financial system, focusing on the wholesale payments network. We estimate that spillovers of an attack on one of the five most active banks would impair 31% of the network, on average, with some days significantly worse. When other banks respond by liquidity hoarding, forgone payment activity can reach more than 2.5 times daily GDP. A reverse stress test shows attacks on groups of small banks can also impair a significant amount of the network. Interconnectedness through third-party providers and financial market utilities poses additional risks.

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1. Introduction

Cyber attacks are an increasing concern especially for financial service firms, which may experience up to 300 times more cyber attacks per year than other firms (Boston Consulting Group, 2019). Almost every financial stability survey includes cyber attacks among the top risks.¹ More recently, academic economists have begun

to highlight the importance of understanding and analyzing cyber attacks as a financial stability risk (Duffie and Younger, 2019; Kashyap and Wetherilt, 2019). Yet, no universal consensus exists on the taxonomy and definition of cyber events, let alone a comprehensive data collection on the frequency and nature of cyber attacks. In this paper, we seek to understand the risk that cyber attacks pose to the U.S. financial system and to quantify how a cyber attack may be amplified through the system.

We begin by integrating cyber risk into the theoretical literature on bank runs, highlighting similarities and differences between cyber and traditional shocks. In some ways, losses related to cyber attacks are similar to other operational loss events that can trigger liquidity runs and lead to solvency issues. Such runs can be

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¹ Survey results come from both industry and regulatory reports. To name a few, see the DTCC's 2018 Systemic Risk Barometer Survey

(<http://www.dtcc.com/systemic-risk/systemic-risk-barometer-surveys>), the Bank of England's Systemic Risk Survey (<http://www.bankofengland.co.uk/systemic-risk-survey/2019-h1>), and the Bank of Canada's Financial System Survey (<http://www.bankofcanada.ca/2019/05/fss-highlights>).

based on panic (Diamond and Dybvig, 1983) or fundamentals (Goldstein and Pauzner, 2005). As in a standard run, a cyber-related run would likely result in direct costs to the affected bank and spillovers to counterparties within the financial sector and to the real economy.

The classification the cyber community uses suggests a taxonomy of how cyber attacks may differ from traditional operational risks (Curti et al., 2019). In a cyber attack, confidentiality of data may be compromised, as may be the availability and integrity of data or systems. Further, a cyber event may be a deliberate attack to damage the financial system. Technological linkages through which cyber attacks can spread are likely to be different from solvency and linkages arising from business interactions. Finally, cyber attacks are likely to be associated with significant uncertainty.

When a cyber attack compromises the availability or integrity of an institution's system and/or data — unlike in a traditional run — it may impair the ability of the bank to service running creditors. In such a scenario, the classic first-mover advantage would be weakened, potentially limiting the incentive to run. On the other hand, uncertainty regarding the nature and extent of the attack could prompt runs to occur in segments of banks' operations that are otherwise unaffected.

In general, rare events are difficult to analyze, as we have seen with financial crises (Gorton, 2014) and macroeconomic disasters (Barro and Ursúa, 2012). Rather than wait to perform a post-mortem analysis of a crippling cyber attack, we instead conduct a pre-mortem analysis to uncover ways that attacks may be amplified, using comprehensive Fedwire data on wholesale payments between financial institutions in the U.S. By doing so, we hope to inform the ex-ante response by mapping out how, in a variety of plausible scenarios, an attack on a single institution or collection of related institutions could be transmitted throughout the financial system. This analysis informs our understanding of transmission mechanisms and the evaluation of vulnerabilities in the financial system, as well as highlighting potential policy responses that would effectively mitigate amplification. In addition, we reverse the analysis to understand different channels through which the system could become impaired.

The wholesale payment network has several characteristics that make it an attractive environment to study. High-value payment and settlement systems may be natural candidates for a malicious attacker intent on inflicting the largest possible damage to the financial system and the broader economy. From a practical standpoint, data on wholesale payments provide detailed information on flows between a diverse set of financial institutions. This detailed information enables us to simulate a broad set of scenarios using actual data and examine the result on the network without relying on too many assumptions. This exercise is similar in spirit to that of Duffie and Younger (2019), but calibrated in the post-Dodd Frank Act environment of ample reserves and heightened liquidity and capital regulation.

We begin by establishing a framework to estimate cyber vulnerability, making simple assumptions about an attack and bank responses, and tracing out the impact

on payments and liquidity. Our framework assumes an attacked institution receives payments but is unable to send payments for a single full day. By accumulating payments from its counterparties, the attacked institution soaks up liquidity, effectively acting as a liquidity black hole. We estimate how the lack of payments leads to liquidity dislocation, and estimate the impact on the liquidity position of other banks that fail to receive payments from the attacked institution. A bank is considered *impaired* if its end-of-day reserves fall sufficiently below the bank's past daily reserve average. We design the framework to be flexible in order to explore various scenarios and variations on assumptions, which illustrate how results may be different depending on the impact of an attack, the responsiveness of banks, and the responsiveness of policy.

In our baseline scenario, we begin by analyzing the impact of a cyber attack on the largest five participants in the network. We estimate that, on average, if any of these large banks stops making payments, 6% of institutions would breach their end-of-day reserves threshold. While the number of affected banks is small relative to the number of banks in the network, the affected banks are often very large. Weighting the impact of a cyber attack by the assets of affected banks dramatically increases the average spillover impact to about 31% of bank assets (38% when including the attacked institution). This finding reflects the high concentration of payments between large institutions, and the large liquidity imbalances that follow if even one large institution fails to remit payments to its counterparties. Disruptions can be even larger within a local area in markets with less bank competition or when a large share of local deposits are held by banks that are more connected in the payment system.

Amplification may also arise when banks connected to a cyber-impaired bank have less capital and liquidity. We estimate that if the liquidity shortfall becomes a solvency shock by turning into disputed payments or interrupted trades, shortfalls could amount to as much as 5% of risk-weighted assets (37% of tier-1 capital) at the 95th percentile of affected banks. At the bank level, impairment is more likely at banks with more unstable funding. Such clustering of different vulnerabilities suggests cyber attacks will be amplified by traditional financial stability vulnerabilities such as leverage and liquidity.

One distinguishing feature of cyber attacks is that they may be designed for maximum disruption. The extent to which an attacker is informed with respect to the payment system, the targeted institution, and its relation to the payment network, may dictate the magnitude of systemic risk arising in an attack. We estimate that attacks targeted on the worst seasonal days for the system are approximately 11% greater than on a typical day, and attacking on the worst date for a particular large institution adds an additional 38% in impairment relative to the case of no specific knowledge.

This initial analysis assumes no response by the banks indirectly affected by the attack. An added impact arises if banks respond strategically to improve their liquidity positions by proactively forgoing payments and hoard-

ing liquidity (a cascade scenario). Theoretically, the effect of liquidity hoarding is unclear, because hoarding can mitigate disruption by bypassing affected banks or amplify disruption by creating new impairments. We find liquidity hoarding amplifies the network impact of the cyber attack, although the effect is relatively modest, by about 1% larger. Although this incremental effect is small, it masks potential disruptions in the system that also arise from a large quantity of payments that would fail to settle as a consequence of strategic hoarding. The breakdown in the settlement of payment activity implies a sizable impact on economic activity even before accounting for spillovers to firms or consumers forgoing spending.

Traditional interconnectedness research examines the network impact of links formed by banks through interbank borrowing and other counterparty relationships. Yet, cyber vulnerabilities may reveal themselves through connections that are not actively captured through traditional counterparty risk management. We explore such vulnerabilities by looking at client lists of significant service providers to banks and by conducting a reverse stress test exercise. For a provider with multiple large and medium-size bank clients, an average of 60% of banks by assets become impaired. For 9% of days, we identify a subset of small domestic banks (under \$10 billion in assets) that can impair at least one of the largest institutions.

Another important consideration is the potential for feedback and amplification through financial market utilities (FMUs). A breakdown in the normal functioning of FMUs that provide liquidity savings, such as Clearing House Interbank Clearing System (CHIPS) and Continuous Linked Settlement (CLS), can dramatically affect liquidity if banks replace those intermediaries with payments through Fedwire. An FMU impairment would require a massive increase in payments value, requiring banks to process additional payments equal to about three times their daily reserves on average.

The paper proceeds as follows. We begin by providing a discussion on cyber risk in the context of the theoretical literature on financial fragility in [Section 2](#). In [Section 3](#), we outline the data and a framework for testing cyber vulnerability. [Section 4](#) summarizes our baseline scenario and main analysis. In [Section 5](#), we augment the baseline scenario by introducing strategic liquidity hoarding behavior of banks. In [Section 6](#), we analyze the systemic implications of cyber risk originating from other non-bank sources, as well as results from a reverse stress test. We consider feedback and amplification to other financial markets in [Section 7](#). [Section 8](#) concludes.

2. Background on cyber risk as a threat to financial stability

We begin by providing background on cyber risk. We then relate cyber risk to the broader theoretical literature on bank runs, highlighting similarities and differences between cyber and other shocks modeled in the theoretical literature.

2.1. Frequency and nature of cyber events

Cyber attacks are rapidly increasing. For example, [ID Theft Resource Center \(2020\)](#) reports that 12,250 breaches were recorded over the 16-year period from 2005 to 2020, and over half of those occurred in the last five years. [IBM Security \(2020\)](#) reports that breaches caused by malicious acts (as opposed to human error or system glitch) are growing even faster, with their share reaching 52% in 2020. Of the malicious breaches in 2020 with a known source, the majority were by financially motivated criminal actors (67%), but the rest were split equally among nation-state actors and hackers and were, on average, at least as costly to the attacked. Although the financial sector had the shortest average time span of breaches, its average time to identify a breach was more than half a year. Further, the lifetime of malicious breaches was, on average, 30% longer than in other industries.

Little information is available publicly on details of specific cyber events, especially in the financial sector. Some notable exceptions are the 2016 “Bangladesh Bank robbery,” in which fraudulent SWIFT messages resulted in the transfer of over \$100 million from the account of the central bank of Bangladesh at the Federal Reserve Bank of New York (see, e.g., [Hammer, 2018](#)). Closely related to this paper, Finastra, a third-party provider of payments services to banks, was targeted in what appears to have been a ransomware attack on March 20, 2020. The firm resolved the attack by disconnecting its servers, leaving certain customers without access payments (see [Finastra, 2020](#)). Finally, details are still incomplete about the SolarWinds hack uncovered in 2020, which potentially affected a range of private sector firms and some U.S. government agencies.

Other recent empirical work in financial economics has begun to quantify cyber risk. [Aldasoro et al. \(2020a\)](#) analyze cyber events across sectors and find that losses are higher when cyber events affect larger firms or multiple firms at the same time, patterns that are very concerning in terms of systemic risk. In addition, they document an inverse relation between the frequency and cost of different types of cyber events, with full-scale business disruptions being the least frequent but most costly. Similarly, they find that in the right tail of the distribution, events with malicious intent have the highest costs, and they specifically highlight cyber attacks on cryptocurrency exchanges.

In a complementary paper, [Aldasoro et al. \(2020b\)](#) analyze a detailed cross-country database on operational risk events in the financial sector (covering 16 years for over 70 large banks), document the evolution of cyber-related events, and compute a cyber value-at-risk (VaR). They find cyber events are still rare but increasing relative to other operational risk events at banks. They estimate that cyber VaR can contribute up to a third of a bank’s total operational VaR.

2.2. Classification of cyber events

Cyber events are conventionally classified into three categories based on the technical nature of the event, introduced in the Federal Information Security Management

Act. However, the classification is also useful for thinking about the economic consequences of a cyber event.

Confidentiality

The confidentiality of data is compromised; for example, customer Social Security numbers or proprietary trading records are hacked and publicized. Such events cause direct losses to the affected bank; for example, the compromise of customer Social Security numbers would incur a loss due to liability for damages, whereas an attack on proprietary trading records might incur a loss due to competitors learning about the bank's strategies. In addition, as for all types of confidentiality events, the bank would suffer reputational costs.

Availability

The availability of data or systems is compromised; for example, a bank's computer systems are shut down. Similar to confidentiality events, availability events can cause direct losses and reputational costs. In addition, availability events can immobilize capital and liquidity, and affect the ability of the bank to perform its core activities. Such events can therefore have considerable spillovers to the bank's customers and counterparties, within and outside the financial sector.

Integrity

The integrity of data is compromised; for example, customer account balances or proprietary trading records are impaired. Similar to confidentiality and availability events, integrity events can have direct costs such as the cost of restoring the integrity of the data and legal costs of resolving issues for which integrity cannot be restored. Similar to availability events, integrity events can have severe spillover costs if they impair the bank's ability to perform its core activities. In particular, integrity events and the resulting legal uncertainty could significantly extend the time required to recover full functioning of the affected bank and the system more broadly.

Based on this classification, cyber events threaten financial stability through their direct costs as well as through the spillovers they cause. Although this classification shows similarity to other types of shocks, cyber events are different along specific dimensions (for a related discussion, see Healey et al., 2018; Curti et al., 2019; Duffie and Younger, 2019; Kashyap and Wetherilt, 2019; European Systemic Cyber Group, 2020; and Adelman et al., 2020).

Intent

A cyber event is typically a deliberate act that may be intended to produce financial gain for the attacker or, more importantly, to create damage to the attacked bank and the financial system. Unlike risk events that impair operations, which may be expected to occur randomly or seasonally, the malicious intent of cyber means that these events may be more likely to happen at times and at banks such that the impact is largest.

Technology

Due to its technological nature, a cyber event can spread through technological linkages, for example, communication networks, as opposed to traditional

economic linkages. In addition, due to the commonality of technology across banks, a cyber event can have significantly larger scale than other types of shocks, affecting a large number of banks at the same time. Combined with malicious intent, attacks with larger potential scale are more likely.

Uncertainty

A cyber event may remain hidden for a considerable amount of time before being detected. The delay in detection increases the potential for damage while undetected and the problems of recovery, especially for integrity events. Even when detected by one bank, other banks may remain uncertain about whether they are affected as well. Again, this characteristic interacts with intent because an attacker may have an incentive to remain hidden as long as possible.

2.3. Impact of a cyber attack on a financial institution

Like any operational risk event, a cyber attack can trigger a liquidity run and lead to solvency issues. Such a run can be based on panic (Diamond and Dybvig, 1983) or fundamentals (Goldstein and Pauzner, 2005). As in a standard run, there are direct costs to the affected bank, for example, due to inefficient liquidation of assets, and spillovers within the financial sector (counterparties) and to the real economy (borrowers).

A detail specific to cyber attacks is that they may impair the ability of the bank to service running creditors, for example, if payments or accounts are not available. In such a scenario, the classic first-mover advantage would be weakened, potentially limiting the incentive to run. This channel through which inaccessibility attenuates the run incentives in the aftermath of a cyber attack, however, relies on a partial equilibrium argument. Because banks and financial institutions operate over a diverse set of markets and perform multiple functions on behalf of clients, uncertainty regarding the magnitude and potential spillover to other segments of the attacked institution may spark runs from clients of other uncompromised operations of the respective bank. This issue is further complicated by asymmetric information problems that arise as attacked institutions formulate their disclosure strategy for communicating with clients.

More generally, cyber attacks have the potential to immobilize capital and liquidity, which can impose costs irrespective of a run or other changes in the behavior of the affected bank and its creditors and counterparties.

2.4. Amplification through network structure and uncertainty

In addition to accounting-based effects (Eisenberg and Noe, 2001), network structure can amplify and propagate shocks (Allen and Gale, 2000). Although these standard models consider the propagation of solvency and liquidity shocks, their insights also apply to cyber shocks. For example, if a cyber shock affects the distribution of or access to liquidity, it can trigger contagion events as in Allen and Gale (2000). The technological impact of a cyber attack may travel through a different network from

interbank lending or payments, as a virus or technical exploit may propagate through data and communications networks, through shared service providers or technological similarities.

In addition, recent theoretical insights derived mainly for solvency shocks also apply to cyber shocks. For example, [Erol and Vohra \(2018\)](#) study the systemic risk in financial networks arising from both correlated defaults of peripheral banks, and defaults of systemically important core banks. Cyber attacks may result in correlated impairments, for example, through technological connections, leading to similar systemic risk.

Asymmetric information plays an important amplifying role in standard theories of systemic risk and also applies to cyber shocks. For example, [Caballero and Simsek \(2013\)](#) show how incomplete information about the location of a solvency shock in a financial network greatly increases individual banks' incentive to engage in individually prudent but systemically harmful actions. This result naturally applies to cyber shocks and banks' individually optimal response in financial networks. Uncertainty about the location of a cyber shock can significantly amplify its disruptive effects. News or rumors regarding impairment of some (possibly unknown) banks or financial institutions could lead to preemptive withdrawals from institutions that are not affected.

Strategic complementarities, coordination problems, and lack of common knowledge have been studied extensively (for a review, see [Morris and Shin, 2003](#)). [Afonso and Shin \(2011\)](#) show specifically for shocks to the payment system — which naturally include cyber shocks — how coordination motives can lead to precautionary demand for liquidity. [Corsetti et al. \(2004\)](#) examines the impact of strategic complementarities with large players, a model that represents well many financial markets and applies to cyber shocks just as it does to fundamental shocks.

A final consideration in cyber is that financial networks structured for resiliency to market risk and for liquidity sharing may be vulnerable to cyber shocks, which connects firms through different networks. If redundancies are not available, networks with a core-periphery structure can see a rapid spread when the core is compromised, and may struggle to identify their connections in the network.

2.5. Policy responses to a cyber event

When thinking about policy responses, it is helpful to distinguish between solvency and liquidity effects that are similar regardless of the initial shock, and those effects that are specific to a cyber-attack related shock. In this way, we can evaluate how ex-ante policy measures to reduce the vulnerability to or the impact of a cyber event, as well as ex-post policy measures in the aftermath of a cyber event, may be necessary. See [Kashyap and Wetherilt \(2019\)](#) for a discussion of ex-ante regulatory and supervisory measures.

Similar to a traditional shock, a cyber event may require ex-post liquidity injections via the discount window, open market operations, or market-wide liquidity facilities. Similarly, ex-ante regulation and supervision can emphasize resiliency to and contingency planning for cyber events.

However, due to the unique properties of cyber events, traditional policy tools such as ex-ante capital requirements or ex-post liquidity provision may not be as effective. For example, a cyber event that falls into the availability or integrity category may require additional policy responses such as a bank holiday to allow for the time necessary to recover the affected systems. In addition, regulatory requirements such as liquidity or reserve requirements could be temporarily suspended if banks are technologically unable to address violations, limiting the knock-on effects of perceived impairment.

More generally, cyber attacks could be mitigated through additional roles of the Federal Reserve or other agencies. For example, the provision of dedicated back-up facilities in core markets could reduce the impact of availability and integrity events ([Duffie and Younger, 2019](#)). Due to the systemic externalities, the individual cost of such facilities would likely outweigh the individual benefit, requiring public provision. In addition, individual banks may not anticipate that, conditional on receiving a cyber shock themselves, other banks have a high probability of being affected at the same time. The implications are similar to that of “crowded trades,” whereby individual banks overestimate the liquidity that will be available when they want to unwind a position. Pre-positioning such facilities and ensuring a cyber attack would not prevent access to them would be critical, because the reduction in uncertainty can be as valuable as the actual facilities. Another element of the “crowded trades” analogy specific to a cyber attack is that many banks may be counting on accessing the same small set of external cyber experts to help them deal with the IT aspects of the attack.

The asymmetric information resulting from a cyber event also creates the potential for beneficial policy intervention. Ex ante, requirements to disclose even minor cyber events to regulators or to share with other banks information on threat assessments and contingency plans could increase resilience by reducing uncertainty and improving collective learning. These disclosures could also boost the credibility of government certification of the information environment. The catalyst for coordination failures is the lack of common knowledge, that is, strategic uncertainty. Ex post, uncertainty could be reduced, for example, by identifying the affected banks or, instead, certifying key banks as unaffected. However, as [Angeletos et al. \(2006\)](#) illustrate, policies surrounding information alone may not suffice to alleviate coordination failures when policies are endogenously determined.

3. Evidence from the U.S. wholesale payment network

In this section, we investigate quantitatively the impact that a cyber attack can have on the financial system, examining how a cyber attack on a set of banks impairs payment activity in Fedwire Funds Service (“Fedwire”). Fedwire represents the majority of wholesale payments between financial institutions in the U.S. and provides detailed information on the accounts and flows between a diverse set of financial institutions. The data enables us to simulate a broad set of scenarios and examine the results

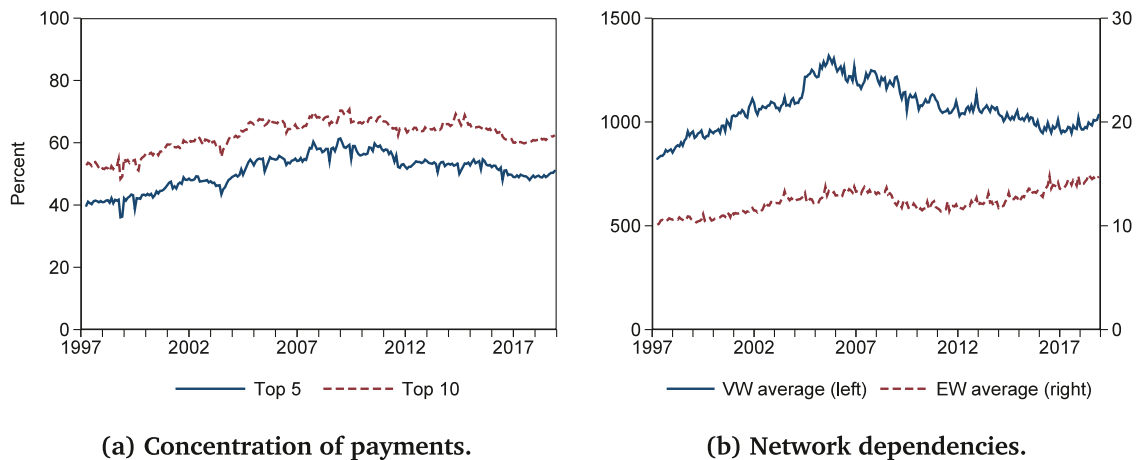


Fig. 1. Concentration and network dependencies. The left panel shows the share of payments sent by the top 5 and top 10 institutions by payments activity. The right panel shows value-weighted (VW) and equal-weighted (EW) averages across participants of the number of other participants receiving payments from a participant.

on the network without relying on heavy parametric assumptions.

3.1. Overview of the wholesale payment network

The U.S. payments network has a core-periphery structure (Soramäki et al. (2007)). Consistent with this structure, payments are highly concentrated among a small set of institutions.

The top five most active banks in the payment system account for close to 50% of total payments, and the top 10 for over 60% (Fig. 1a). Activity is concentrated not only in terms of payments value, but also in terms of network connections, with the most active banks' connections outnumbering the average banks' connections by several orders of magnitude (Fig. 1b).

The dramatic change in the underlying reserve environment after 2008. Past studies that highlight strategic behavior in liquidity management, both empirically and theoretically, assume a system with scarce reserves. Following the financial crisis, the Federal Reserve's asset purchasing programs expanded its balance sheet, and with it, the aggregate quantity of reserves. As a result, the turnover (the value of payments made divided by total liquidity) has dropped considerably (Garratt et al., 2014).

Although a complete analysis of the changes to the underlying payment environment is beyond the scope of this paper, we provide some context and relevant dimensions to understand the analysis of cyber attacks.

Existing empirical literature documents strategic behavior and complementarities in the wholesale payment system. When liquidity is scarce, banks manage intraday liquidity by delaying payments and closely matching inflows and outflows to avoid liquidity shortages (McAndrews and Rajan, 2000). Over the normal course of the day, this creates both clusters and delays in the aggregate value of payments. Klee (2010) shows that disruptions due to opera-

tional problems increase aggregate uncertainty on market functioning and are realized in prices, volume, and discount window activity. Afonso and Shin (2011) simulate intraday liquidity dynamics and show network externalities in liquidity-hoarding behavior can spark illiquidity spirals in payment systems.

Past shocks offer a glimpse into how the payment network behaves in times of crisis. On September 11, 2001, many banks based in lower Manhattan experienced direct damage to property and communication systems (Lacker, 2004), which were followed by payment coordination breakdowns (McAndrews and Potter, 2002). At the time, the Federal Reserve was able to use large liquidity injections to restore coordination in the payment network and limit any knock-on effects. If the only impact of a cyber attack is a liquidity shortage, the Federal Reserve can provide liquidity as in past crises, as long as compromised institutions are solvent and have prepositioned collateral. However, a cyber attack is likely to be associated with significantly greater uncertainty about the effectiveness of any policy response and can also affect private incentives to hoard liquidity in unexpected ways. Ashcraft et al. (2011) find evidence of strategic liquidity management in interbank markets during the 2007–2009 financial crisis. In recent work, Allen (2020) considers the impact of operational risk events more broadly on the payments system.

Some measures of banks' strategic management of intraday liquidity have fallen with the abundant supply of reserves. For example, delays in payments and turnover, metrics that proxy for strategic complementarities in payment activity, have dramatically decreased (McAndrews and Kroeger, 2016). However, other post-financial crisis changes, such as liquidity regulation, liquidity stress testing, and improved liquidity risk management, mean banks continue to closely manage their liquidity. More recently, these proxies for strategic complementarities have shown signs of increasing again, suggesting that even in an environment with an ample quantity of

reserves, banks' payment activities continue to be carefully managed.

3.2. A framework for cyber vulnerability

Our main strategy is to analyze scenarios in which a cyber attack impairs a subset of institutions, and to evaluate the systemic implications by examining the overall impact on the payment network. Throughout the paper, we focus our analysis on a single-day shock to the payment network. Multi-day impairments necessarily result in a broader impact, as we show in Section 4.6. To fix ideas, we outline a framework used to understand how hypothetical cyber attacks would be amplified through the system.

As highlighted in Section 2, a cyber incident can compromise normal functioning of an institution through affecting a bank's confidentiality, availability, and/or integrity. We focus on scenarios involving availability or integrity. A cyber attack may temporarily impair the availability of relevant data or communication and messaging systems of a target institution. For the duration of the issue, the institution may not be able to process or transmit payments. Alternatively, a cyber attack may compromise the integrity of a target institution's data, either by manipulating or corrupting the data. Although the target institution could, in principle, make payments on behalf of its clients and/or its key operations to avoid reputational costs, and/or delay announcing its compromised position, this decision would be unlikely, given the large notional size of payments and because payments in Fedwire are settled with finality. The alternative may be that the institution undergoes a process of reconciliation in which it rebuilds its data both internally and externally. In either case, a successful cyber attack can have an immediate and direct impact on an institution's ability to carry out day-to-day operations.

A hypothetical scenario starts with a set of target institutions that experience a cyber attack, and are able to receive but unable to remit any payments over the Fedwire operating day. The assumption that only the sending of payments is affected reflects the institutional practices of Fedwire. In brief, a payment is actualized when Fedwire receives a payment request from the sender. As a result, an institution's balance, as represented in the Fedwire system, can immediately reflect changes due to incoming payments, even if the institution is unable to observe or interact with the Fedwire network due to a cyber incident. Attacked institutions can therefore become liquidity black holes, soaking up liquidity but unable to send out payments (in the form of reserves). The inability to send payments can restrict the effective flow of liquidity, similar to the situation after September 11, 2001 (Lacker, 2004).

Let U be the universe of financial institutions, indexed i , in the network. A scenario S specifies three primitives (Z, R, B) , where $Z \subset U$ is the set of financial institutions that are directly affected by the cyber attack (referred to as *shocked* institutions), R is the set of reaction functions of institution $i \in Z^c = U \setminus Z$ that characterize its payment behavior within a given day, and $B = \{b^i\}$ is the set of *impairment* conditions whereby an institution i is considered adversely affected by the scenario if condition(s) b^i holds.

With a slight abuse of notation, we also use B to denote the set of institutions $i \in U$ that are impaired. By default, we assume any directly affected institution $i \in Z$ is also impaired. As such, an outcome used to characterize the network impact includes both that of the *primary* impact (via institutions $i \in Z$) and the *derivative* impact (via institutions $i \in Z^c$).

A scenario provides us with a baseline to describe the behavior of the constituents of the network and evaluate whether an institution is impaired by the scenario. Fixing a scenario, we first simulate the scenario for the entire sample period, and construct a dataset with counterfactual outcomes, including the list of impaired institutions. We then assess the effect of the shock by evaluating and analyzing the properties and characteristics of impaired institutions implied by the counterfactual outcomes. We focus on three different aspects. First, what is the magnitude of the network impact caused by the scenario? Second, what is the set of institutions that are vulnerable to the shock? Finally, how does the level of information available to the attacker relate to the systemic implications of a cyber attack?

3.3. Data description

Our primary data sources consist of intraday payment data between banks in the Fedwire Funds Service payment system and the individual banks' end-of-day reserve balances over the year 2018. During this period of time, the aggregate supply of reserves gradually shrinks, with depository institutions collectively holding about \$2.3 trillion in reserves in January 2018, and about \$1.8 trillion in reserves at the end of 2018.²

Table 1 summarizes statistics on daily payment value. The average total daily payments in Fedwire in 2018 is about \$2.8 trillion, with a standard deviation of \$320 billion and a maximum of over \$3.7 trillion. The cross-sectional variation is considerably larger with a highly skewed distribution of total payments sent at the institution level. The median bank sends less than \$1 million per day, whereas the average bank sends about \$500 million per day and the most active banks send over \$7 billion per day.

Fedwire payments data are organized by "ABAs" — account numbers instituted by the American Bankers Association to facilitate check payments. Because some banks have multiple ABAs (typically due to mergers), we aggregate entities to the parent depository institution level.³ We use data from Call Reports, Y-9C and FFIEC 002 filings, as well as Summary of Deposits data, to complement Fedwire data with detailed information on financial institutions, in-

² Although the aggregate quantity of reserves and distribution of reserves throughout the system will affect this analysis, daily variability is sufficient to inform the analysis.

³ Our analysis is at the depository institution level for two reasons. First, financial institutions may face liquidity demands and regulations, including reserve requirements, at the depository institution level. Second, impairments in the payment system could make shifting liquidity between accounts difficult, even within the same bank holding company. Our main results continue to hold if we aggregate accounts to the bank holding company level.

Table 1

Summary statistics.

The table shows summary statistics of daily payments value in Fedwire.

	Avg.	St. dev.	p1	p25	p50	p75	p99
Total sent _{<i>i,t</i>} (millions)	519.94	9588.86	0.00	0.00	0.37	3.16	7242.90
Avg. by institution <i>i</i> (millions)	508.75	9318.75	0.00	0.15	0.86	4.29	7818.85
Total on day <i>t</i> (trillions)	2.85	0.32	2.32	2.63	2.81	3.02	3.73

cluding asset size, branch location, and other relevant information. As we show, taking into account institutions' size is integral to understanding and evaluating the overall impact of a cyber attack.

Thus, our main universe *U* consists of financial institutions for which information on asset size is available, which still permits a wide class of U.S. chartered financial institutions, including national banks, state member and non-member banks, federal and state savings banks, and branches of foreign banks. These institutions together represent about half of the participants in the Fedwire payment system and account for about three quarters of total daily payments and reserves. All reported results on the network impact are based on this subset of institutions. We exclude all other institutions from the analysis of network impact. By doing so, our results on both the unweighted and weighted impact of cyber attacks are based on the same subset of potentially impaired institutions. In addition to excluding a number of non-depository institutions, many institutions excluded from our analysis are credit unions, which together represent about 18% of participants but collectively account for less than 0.5% of payments by value. By excluding these institutions from *U*, we are potentially underestimating the network impact, because excluded institutions may be heavily affected but are not included as potential firms in *B*. Second, as we show in Section 5, excluded institutions are implicitly assumed to continue to send out payments unconditionally, which underestimates the cascade effect.

4. Baseline scenario

In this section, we outline our baseline scenario and summarize the results. In the baseline scenario, we consider an attack on a single institution on day *t* and assume the attacked institution does not send any payments on that day. All other institutions are assumed to make payments as observed in the payments data for day *t*. We calculate the counterfactual end-of-day-*t* reserve balance for each institution and analyze the set of institutions *B* that drop below an impairment threshold (Section 4.1). Because only the shocked institution's payment activity is assumed to be disrupted, the impact in the baseline scenario arises exclusively from other institutions failing to receive payments from the shocked institution. We consider the impact of other institutions endogenously suspending payments in Section 5. Several rationales exist for this simple baseline scenario. First, some banks may not have the systems or protocols in place to identify and respond immediately to the failure of a counterparty to make payments. Second, banks may not be sensitive to intraday liq-

uidity conditions, due to the ample reserves environment, and thus may not react to abnormal conditions in a timely manner. Finally, the baseline scenario may also be interpreted as payment network participants continuing to operate as usual in expectation of a regulatory intervention.

We focus attention on attacks on the most active institutions in the payments system. As described in Section 3.1, a well-documented and stable feature of the U.S. wholesale payments system is its high concentration of activity among a small set of financial institutions. The core banks in the network are among the largest banks in the U.S., with a rank correlation between assets and payments of over 80%, and therefore play vital roles in various other markets, including Treasury, equity, and derivatives markets. This feature of the payment system strongly suggests that a cyber attack at a single institution alone could materially impair normal functioning in the payment network. We therefore present results for a hypothetical situation in which *Z* consists of one of the largest five institutions in the payment network ("top-5"), by value in 2018.

4.1. Defining impairment

An important unknown in evaluating the severity of a cyber event is under which conditions an institution that is not attacked directly becomes materially impaired. For instance, even though we observe each individual bank's reserve levels at a high frequency, we do not directly observe the bank's desired reserve holdings, nor do we observe levels at which the bank would view its liquidity position as severely compromised. For institutions subject to regulatory requirements such as the liquidity coverage ratio (LCR), high-frequency and detailed data on balance sheet are needed to assess how binding liquidity regulation is for an institution. Furthermore, banks' internal liquidity targets may incorporate a buffer above the regulatory minimum. Additional liquidity may be required within entities of the bank holding company to support recovery and resolution planning. Because individual reserve balances are endogenously determined by both market-wide liquidity conditions and each bank's idiosyncratic liquidity demand, a bank may manage its day-to-day reserve balances to accommodate shocks to its balance sheet positions.

One way to approximate whether a bank's liquidity position is compromised is to use empirical variation in banks' reserves as a basis for evaluating when they would be materially impaired. Our main measure uses individual banks' end-of-day reserve dynamics to identify when a cyber event has a significant effect on a bank's liquidity position. In our baseline definition, we classify an institution as *impaired* if the counterfactual end-of-day reserve balance

r_t^i is more than two standard deviations below its average, over a 30-day window⁴:

$$b_t^i = \bar{r}_{t-1,t-30}^i - 2\sigma_{t-1,t-30}^i \quad (1)$$

A key advantage of our reserve-based impairment measure is that it provides a time-varying cutoff that adjusts as individual banks optimize their reserve balances to their liabilities and liquidity positions as well as to changes in aggregate reserves.

How binding is b_t^i ? One way to answer this question is to examine how close banks come to their impairment threshold during normal times. Consider the normalized distance to the past 30-day average given by a time-varying z-score measure:

$$z_t^i = \frac{r_t - \bar{r}_{t-1,t-30}}{\sigma_{t-1,t-30}} \quad (2)$$

On average, the z-score is negative most of the days, with a median of -0.03 . There is significant positive skew with a mean of 0.49 . Table A.4 in Appendix A summarizes the average z-scores for banks, weighted by size. The table shows that banks, on average, stay within roughly half a standard deviation above and below their trailing average balance. As reference, Fig. A.11 in Appendix A shows the time-varying buffer relative to reserve balances for the top-5 institutions. Overall, this finding suggests banks, adjusting for size, tend to actively manage a relatively stable end-of-day reserve balance.

While our current impairment measure is chosen for its intuitive, parsimonious definition, the framework is designed to be flexible to different definitions of impairment b_t^i . The impairment definition can be adjusted to be more or less stringent, simulating possible policy responses to an attack. For example, if reserve balances reflect compliance with liquidity regulation or supervisory expectations, a policy response could be to change those regulations or expectations in order to mitigate the amplification of an attack.

4.2. Distribution of impact across institutions and over time

As a starting point, we examine the share of institutions that become impaired, that is, end-of-day reserves that fall below their impairment threshold, as a consequence of a successful cyber attack on one of the top-5 institutions. Fig. 2a outlines the average daily impact of an attack on each of the five institutions in 2018, where unweighted share is the fraction of impaired institutions, that is, $\frac{|B|}{|U|}$. Across all days in 2018, the average share of institutions impaired ranges from 4.8% to 8.5%, depending on which of the top-5 institutions is attacked (blue bars). We see substantial variation on different days of the extent of amplification of an attack on a single large institution. The daily variation in the share of impaired institutions is

larger within each shocked institution than it is across the shocked institutions. The share impaired on the 1% worst days is roughly three times larger than the share on the 1% best possible days (dashed whiskers).

By comparison, we also examine a size-weighted share of the impaired institutions, which is computed using $\frac{\omega_q^i |B|}{|U|}$, where $\omega_q^i = \frac{A_q^i}{\sum_q A_q^i}$ and A_q^i is the size of the institution measured in total assets in the quarter q (red bars in Fig. 2a). Across the top-5 institutions, weighting impaired institutions by asset size leads to between 22% and 55% of bank assets being impaired, magnifying the average impact by over four times relative to the numerical share of affected institutions.

Fig. 2 b shows histograms of the distribution across days in 2018, averaged across the shocked top-5 institutions. Compared to the unweighted distribution (blue), the weighted distribution is shifted to the right and exhibits a more symmetric form, whether the shocked top-5 institution is counted among the impaired or not (red or gray, respectively). Because the directly shocked institutions are large in assets, the direct impact of the shocked institution has a nontrivial contribution to the total network impact. However, this direct effect does not explain the variation in the simulated network impact. Even excluding the shocked institution, on the days of maximum impact, an attack results in close to half of the network being impaired, in terms of bank assets. This distribution is shown in the gray bars of the histogram, which nets out the direct impact on the top-5 institution. The large amount of assets impacted reflects the high concentration of payments between relatively large institutions, and the large liquidity imbalances that follow if even one large institution fails to remit payments to its counterparties. Because of the high volume and size of payments among core institutions, large and highly central institutions are more commonly affected by an attack on any of the top-5, and represent a large share of the affected banks. In other words, the large share of affected assets generally reflects transmission from one top-5 bank to another.

Of course, different assumptions would produce different estimates of the extent of amplification. In terms of comparative statics, any ability of the attacked institution to send out some payments would result in less amplification. Similarly, lowering the impairment threshold would also reduce amplification. Changes to the information environment also affect amplification. For example, if banks were informed about the impairment and stopped sending payments to the affected institution, their liquidity would also be improved. We explore the impact of allowing for different responses from other banks in Section 5.

4.3. Geographic impact

We began by estimating the number and assets of affected banks on a national basis. A key concern in a cyber attack is that disruptions in the wholesale payment network can lead to widespread spillovers to the real economy. In this section, we enrich our analysis of amplification from a single direct attack in our main scenario, by understanding how the physical structure of the U.S. bank-

⁴ Because the impairment threshold b_t^i is a moving average based on past reserves, it is possible that a bank's end-of-day reserve balance drops below b_t^i on certain days, even before accounting for missing payments from our scenario. We show in Appendix B that excluding from the analysis any banks that are impaired but do not receive any payments from the shocked institution on that day does not materially affect our results.

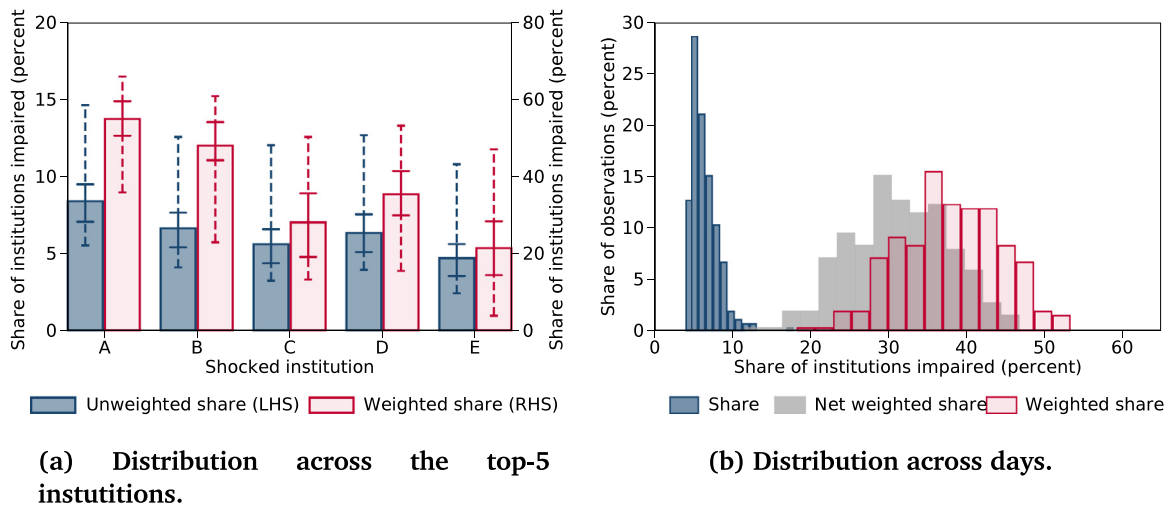


Fig. 2. Impact of a shock to a top-5 institution. The figure shows the distribution of the unweighted share (blue), share weighted by assets (red), and net weighted share (grey) of institutions impaired by a shock to each of the top-5 institutions. Panel (a) shows the distribution across the top-5, averaged across days; bars represent the average impact; solid whiskers represent the p25/p75 range; dashed whiskers the p1/p99 range. Panel (b) shows the distribution across days, averaged across the top-5 institutions. Net weighted share excludes the shocked institution. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ing system could transmit shocks to bank depositors and customers. Given the high concentration of activity in key financial centers in the U.S., a natural question is, to what extent a successful cyber attack on a large systemic institution has an impact on particular geographic areas.

To examine the geographic impact on the banking sector, we evaluate the overall market share of impaired institutions by geography in two steps. First, for each institution $i \in U$, we make use of Summary of Deposits data to measure deposit market share in each respective state, metropolitan statistical area (MSA), and county, which proxies for banking market presence. Then, for each scenario, we aggregate the overall impact by total market share of the set of impaired institutions B .

To the extent that banking markets are concentrated, an impairment of a small number of institutions may have a large effect on depositors in that market, particularly when those institutions are highly connected in the wholesale payments network. We find a significant amount of variation in the extent of an impact. Fig. A.12 in Appendix A depicts the geographical impact aggregated at the county level. Geographic impact varies because the extent to which the shock affects banking markets depends not only on how connected relevant institutions are to the network, but also on how concentrated markets are. For instance, Ohio sees, on average, 53% of its banking sector impaired under the main scenario, because it has a high concentration of banking activity by banks within the core of the payment network.

Examining the top 10 MSAs with the largest shares of deposits held by impaired institutions, in the baseline scenario, we find all 10 MSAs average at least 40% of the local banking market impaired. Although the geographic impact is correlated with MSAs with greater financial activity, the impact is also large for MSAs where the financial sector does not play an outsized role in the local economy. This result can be attributed to a combination of two factors:

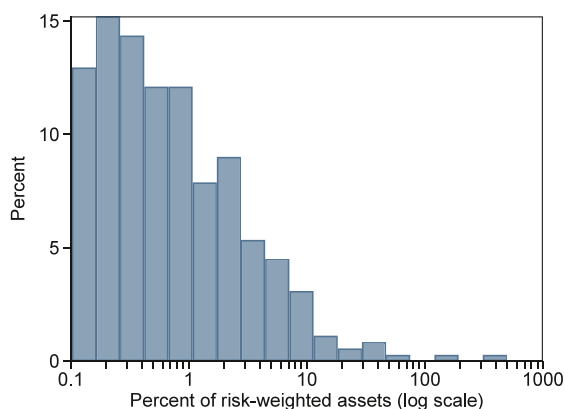
concentration in the banking market, and the deposit market share in the MSA of core institutions in the payment network. The impact is even more extreme in some counties where the banking market is highly concentrated.⁵

This result illustrates that disruptions in wholesale payments and subsequent liquidity dislocations can potentially prompt widespread illiquidity of banks that can be very concentrated in certain geographies. Notably, a cyber attack can lead to uncertainty regarding the security and safety of the banking system that can be felt by depositors and financial market participants, both at the institutional and individual household level. This finding points to a broader concern: a cyber attack has the potential to trigger rapid degradation in the trust towards financial institutions and can therefore quickly lead to traditional forms of financial fragility. At the micro level, this finding also shows an additional possible risk of highly concentrated banking markets. Disruption in a geographic area is likely to be higher if a large share of customers and businesses cannot access their deposits. Although a lack of reserves may not immediately lead to a lack of cash in ATMs, if affected banks delay payments or reduce lending, the impact is likely to be magnified in some geographies.

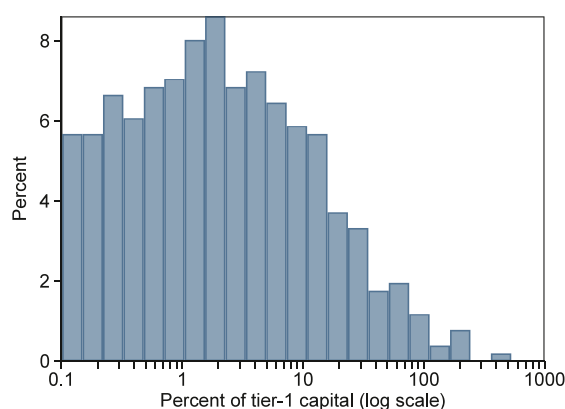
4.4. Solvency and stability impact

Our discussion of impact is mainly in terms of banks becoming impaired, where a bank's impairment is defined in Section 4.1 as its reserve balance falling more than two standard deviations below its trailing average. We now consider to what extent impairment presents an explicit

⁵ Note that the Summary of Deposits data have not been adjusted to account for the way in which some corporate and online deposits are accounted for at bank headquarters locations, meaning that measures of concentration in MSAs where the large banks are headquartered may be overstated.



(a) Shortfall relative to risk-weighted assets.



(b) Shortfall relative to tier-1 capital.

Fig. 3. Distribution of liquidity shortfall across impaired banks. The figure shows histograms of the shortfall in reserves relative to risk weighted assets (Panel a) and relative to tier-1 capital (Panel b). Averaged across shocks and days, truncated to values between 0.1% and 1000%.

solvency risk to banks and how impairment is related to banks' other vulnerabilities and therefore amplifies financial stability risks.

We consider the potential direct solvency implications of our baseline one-day scenarios of a cyber-attack compromising one of the top-5 banks. For each of the banks that becomes impaired due to not receiving payments, we calculate the liquidity shortfall, that is, the difference between their actual end-of-day reserve balance and their counterfactual end-of-day reserve balance calculated in our shock scenario.

Figure 3 shows distributions of the liquidity shortfall across impaired banks, scaled by risk weighted assets (Panel a) and by tier-1 capital (Panel b). This exercise is similar to the payments never being made to the bank, and the loss of payments being realized as an accounting loss, as may occur if the books and records of a bank are permanently impaired. Alternatively, this measure could be interpreted as a proxy for counterparties' assessment of the exposure to an impaired institution. Note, the liquidity shortfall measures differ from typical exposure measures because they capture the exposure of banks' customers as well as the direct exposure of the bank. The magnitudes are large when compared with risk-weighted assets or tier-1 capital. Considering the right tail of the distribution across banks (95th percentile), the shortfall amounts to as much as 5% of risk-weighted assets or 37% of tier-1 capital.

The financial stability implications more broadly may depend on banks' other vulnerabilities, because the uncertainty around an attack is likely to have more severe implications for banks with less liquidity or less capital. Thus, we regress the fraction of days a bank becomes impaired in one of our top-5 scenarios on the bank's size, liquid asset share, its unstable funding share, and capital ratio.⁶ Table 2a shows the results and Table A.5 in Appendix A provides summary statistics of the variables.

Consistent with our results in Section 4.2, we find larger banks are significantly more likely to become impaired ($p\text{-val} < 0.01$, columns (1)–(5)). As previously noted, this poses a serious financial stability concern as stress at larger banks has more systemic consequences. Controlling for size, we find impairment is significantly related to more unstable funding ($p\text{-val} < 0.05$, columns (3) & (5)). The implications on large banks is a further financial stability concern if we consider wholesale funding sources more likely to withdraw from an impaired bank. Potentially mitigating this instability, we find impairment is also related to higher capital ($p\text{-val} < 0.01$, columns (4) & (5)) and more liquid assets ($p\text{-val} < 0.01$, columns (2) & (5)). Across the four explanatory variables in the full specification in column (5) of Table 2a, bank size has the largest effect with a one-standard-deviation increase in log assets increasing the fraction of days impaired by 2.1 percentage points (compared to a baseline of 7.4%). Effects of the balance sheet ratios are half as large, with a one-standard-deviation increase in the liquid asset share by 1.0 percentage point, unstable funding share by 0.8 percentage points, and capital ratio increasing the fraction of days impaired by 1.0, 0.8, and 1.4 percentage points, respectively. Table 2b reports the same specification but restricting the sample to progressively larger banks. Results are similar in this sub-sample, but the mitigating factor of more liquid assets is no longer significant (columns (3)–(5)).

4.5. Timed attacks

A defining feature of cyber risk is that attacks may be targeted, with the intent to achieve maximum disruption rather than financial gain. The extent to which an attacker is informed with respect to the payment system overall and the targeted institution specifically can affect the magnitude of systemic risk associated with a cyber attack.

⁶ Unstable funding is defined as non-retail deposits, fed funds and repo borrowing, commercial paper, trading liabilities, and other borrow-

ing with maturity less than one year. The unweighted capital ratio is tier-1 capital divided by total assets.

Table 2

Relating impairment to bank characteristics.

The tables show estimates from linear regressions at the bank level of fraction of days impaired (averaged across shocks) on the listed controls. Panel a pools all banks; Panel b restricts each regression by bank size as noted at the top of each column. $p < 0.1$, $** p < 0.05$, $*** p < 0.01$.

(a) Individual variables					
	Fraction of days impaired				
	(1)	(2)	(3)	(4)	(5)
Log assets	0.009*** (0.002)	0.011*** (0.002)	0.011*** (0.002)	0.011*** (0.002)	0.015*** (0.002)
Liquid asset share		0.143*** (0.032)			0.094*** (0.034)
Unstable funding share			0.049** (0.022)		0.052** (0.023)
Capital ratio				0.357*** (0.074)	0.326*** (0.078)
Adj. R2	0.006	0.013	0.007	0.014	0.019
Num. obs.	2688	2688	2688	2688	2688
(b) Splits by bank size					
	Fraction of days impaired				
	(1) All banks	(2) > 1 bn	(3) > 5 bn	(4) > 10 bn	(5) > 50 bn
Log assets	0.015*** (0.002)	0.041*** (0.004)	0.056*** (0.011)	0.084*** (0.014)	0.152*** (0.035)
Liquid asset share	0.094*** (0.034)	0.170** (0.066)	0.253 (0.178)	−0.225 (0.221)	−0.045 (0.381)
Unstable funding share	0.052** (0.023)	0.127*** (0.044)	0.478*** (0.109)	0.713*** (0.134)	0.709*** (0.219)
Capital ratio	0.326*** (0.078)	0.474** (0.223)	0.493 (0.608)	2.843*** (0.813)	4.146** (1.864)
Adj. R2	0.019	0.167	0.321	0.476	0.554
Num. obs.	2688	627	178	118	37

An important dimension of private information entails the timing of a cyber attack. Although payment volumes are high throughout the year, utilization varies depending on the demand for transactions by both the real and financial sector. The most active days see over 60% more total payments than the least active days (see Table 1). Perhaps unsurprisingly, these large swings in payment value also correspond to large variation in the network impact of a cyber attack. Even within the top-5 institutions, shares of total payments are variable, and result in large swings in the transmission of shocks to other banks over the course of the year.

The variation in total payment value over time indicates that the timing of the cyber attack can heavily influence the magnitude of impact. In this section, we evaluate how progressively higher levels of knowledge that a cyber attacker may possess increase the attacker's ability to inflict damage to the payment network, and to what extent information can be effectively used to disrupt financial activity.

Public information

The first level is public information. Past studies document strong seasonality in wholesale payment activity, with large, periodic increases in payment activity over the course of a year (Klee, 2010; Furfine, 2000; Soramäki et al., 2007). A cyber attack may therefore be timed to coincide with high-activity periods, so as to increase the network impact. To illustrate this relation, we examine the counterfactual impact of a subset of days coinciding with

calendar periods documented by existing studies to have higher than usual payment activity (referred to as “seasonal” days), and compare it with an average Fedwire day in 2018. Timed attacks on seasonal days increase the weighted impact by about 11% on average.

Private information on a target institution

This second level is private information, specifically pertaining to a target institution. A common form of cyber attack is an “advanced persistent threat” (APT), during which the attacker is undetected and may extract data on an institution's activities and obtain information on the internal mechanisms within the institution. For example, Mexico's Bancomext and Chile's Bank of Chile fell victim to malicious actors who gained access into their systems but remained in stealth mode for extended periods of time before exploiting system weaknesses.⁷ These instances primarily involved thefts in the order of millions of dollars. However, given a different objective, such as inflicting the largest damage, an attacker could attempt to analyze the compromised institution's financial and payment activity. In the same vein, private information and insider knowledge may become available via direct and indirect forms of insider threat (Randazzo et al., 2005).

In the context of payment networks, a valuable type of private information is understanding patterns in the in-

⁷ <https://www.wired.com/story/mexico-bank-hack/>

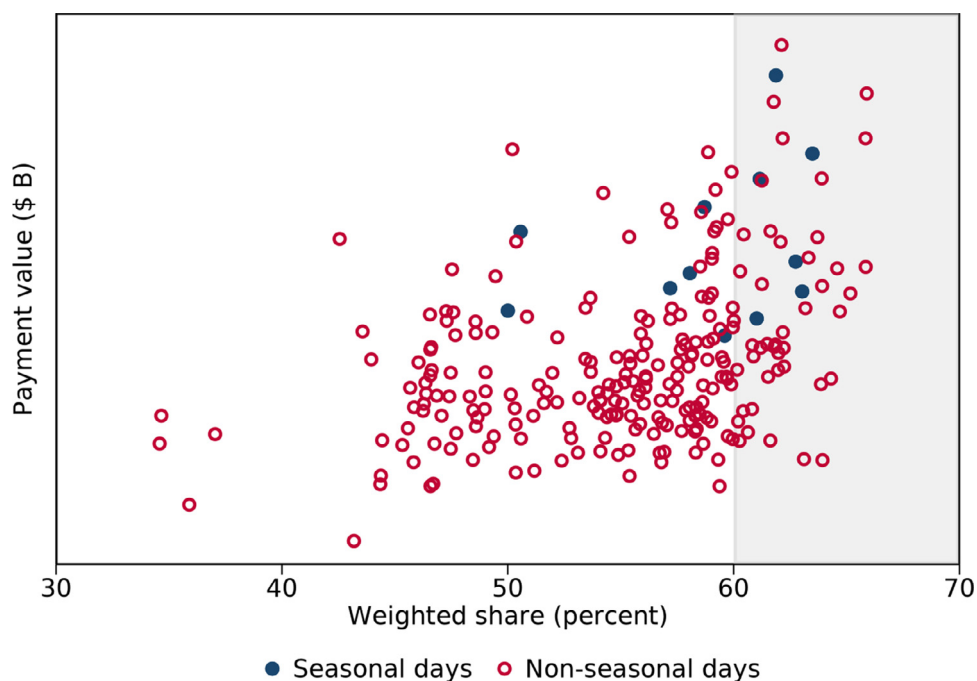


Fig. 4. Relation between payment value and network impact. The figure shows a scatterplot of daily payments sent by one top-5 institution against the weighted share of impaired institutions from the baseline scenario of the same top-5 institution.

dividual institution's payments. As hinted earlier, days in which an institution conducts large payments are correlated with days in which the scenario yields a large share of impaired institutions. Fig. 4 plots the weighted share of impaired institutions on a particular day against one top-5 institution's total payments on that day. As expected, we see a positive correlation between the two measures. Days with a high impact include many days that are seasonal days as well as days on which the attacked institution has high payment value.

For each of the five institutions, we identify days on which it individually has high payment value in 2018. Attacks on days with high payment value translate into a greater network impact relative to seasonal days, and increase the network impact relative to the average day by about 25%.

Private information on network interconnectedness

What is the largest possible shock that can be inflicted on the network? This ex-post identification of worst-case days proxies for circumstances under which an attacker is privy not only to private information regarding a target institution's payment activity, but also regarding the underlying network and counterparties' activities. The average of the maximum impact for a top-5 institution is considerably larger than that of an average day, with about a 63% increase in network impact. Although access to such detailed information may be unlikely, it provides us a way to estimate how much access to private information can magnify the systemic consequences of a cyber attack, as well as serving as a reverse stress analysis.

In all of these examples, we show the extent to which the attacker is privy to detailed institutional and insider

knowledge on the payment network, banks' relations, and banks' operations can substantially heighten the systemic risk associated with a cyber attack. One limitation of the payment network data is that we are not able to directly observe the purpose of these payments. An attacker that is able to extract further information on payments, such as the payments' purpose or the identities of underlying bank customers, could achieve greater disruption to the economy, for example, by timing disruptions to coincide with payroll dates or political events such as elections.

4.6. Multi-day attack

Our baseline scenario provides a lower bound for an attack that persists over multiple days. A cyber attack that shuts down critical infrastructure for an extended period of time may generate even larger negative spillovers to the economy (Council of Economic Advisers, 2018). In the context of financial stability, an extended disruption to key institutions or FMUs could be particularly destructive because banks' incident response plans may not be designed to combat long-term disruptions. There are several challenges to assessing the multi-day impact of a cyber attack. A cyber attack with longer-term disruptions is likely to be met with actions from various stakeholders, including regulators and central banks. Like any shock that leads to illiquidity, a central bank may be able to counteract transient illiquidity episodes.

Understanding these limitations, we extend our analysis to evaluate the multi-day impact of a cyber attack in Appendix C. If the attack continues beyond the first day, we estimate that the shock transmits to more core institu-

tions, with the average size of impaired institutions reaching \$85 billion by the second day, and over time, affecting other periphery banks. The average liquidity shortfall in the system grows from \$122 billion on the first day, to an average of \$1 trillion by the fifth day. This amount is roughly equivalent to 20% of the Federal Reserve balance sheet during our sample period, signifying the magnitude of liquidity injections that could be required if a cyber attack impaired the data and records necessary to make payments.

5. Liquidity hoarding and network cascades

The baseline scenario assumes all institutions other than the directly attacked institution continue to make payments as usual, even after they fail to receive expected payments from a core institution. To study a richer strategic interplay, we now consider an augmented scenario, which we refer to as the *cascade scenario*, in which institutions react to abnormal imbalances in their intraday reserve positions.

The absence of payments from an affected core institution is likely not only to create large liquidity dislocations at the end of the day, but also to have a significant impact on the intraday balances of counterparties. Relative to an operational outage, a suspected cyber attack may be accompanied by more uncertainty and lack of common knowledge regarding the source, magnitude, and implications of the attack. Given the malicious intent of a cyber attack, market participants may also be concerned about extended disruption and delayed resolution. Uncertainty regarding the nature of a suspected cyber attack may also be sustained endogenously — target institutions may be reluctant to disclose to counterparties and clients the exact state of their internal systems, and may delay communications until they acquire enough information. Such delays can also contribute to asymmetric information, prompting institutions to stall payments and attempt to retain as much liquidity as possible (in risk of operational default). Finally, even if a particular bank is happy to send payments, its customers may hoard balances or try to effect payments through other means (Duffie and Younger, 2019).

5.1. Intraday reaction functions

An institution may take notice of abnormal payment activity if it observes unusually large deficits in its intraday liquidity position. Let $\eta_{\tau,t}^i$ be the net payment deficit of institution $i \in U$ on day t up to minute τ (total outflows minus total inflows). We assume each institution i uses a trigger strategy $r^i \in R$ that characterizes its intraday reaction to changes in its net payment deficit: if $\eta_{\tau,t}^i$ surpasses some threshold, institution i stops sending payments for the rest of the day. In our main specification, we take this threshold value to be the maximum realized net payment deficit of the institution in the entire year of 2018. This threshold, denoted $\eta_{\max}^i$, corresponds to the maximum difference that an institution had between its payment inflow and payment outflow under normal operations, at the minute level. Bank i 's reaction function is then given by

$$r^i(\{\eta_{\tau,t}^i | \tau \leq k\})$$

$$= \begin{cases} \text{send payments at minute } k & \text{if } \eta_{\tau,t}^i < \eta_{\max}^i \text{ for all } \tau \leq k \text{ on day } t \\ \text{hoard liquidity} & \text{otherwise.} \end{cases} \quad (3)$$

There are two different channels that may trigger a bank to hoard liquidity. The primary channel is due to deficits that grow as the institution affected by the cyber attack fails to remit payments that it would have remitted under the normal course of the day. When the volume of such payments to institution $i \in Z^c$ becomes sufficiently large, institution i may halt its own payments in order to preserve and absorb incoming liquidity, in anticipation of aggregate illiquidity at the end of day. Thus, the trigger strategy r^i can be interpreted as the reaction to an informative signal (i.e., $\eta_{\tau,t}^i$) through which institution i becomes aware of the possibility of a cyber attack on a large systemic institution, or as an optimal liquidity management strategy to maximize the probability that its end-of-day reserve balance does not fall below its internal target, that is, $r_t^i > b_t^i$.⁸

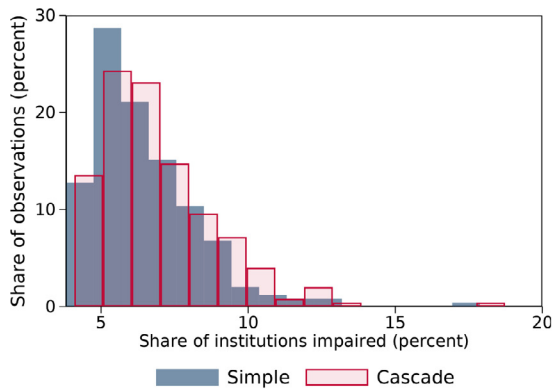
A secondary channel that would trigger a bank to hoard liquidity arises from illiquidity spillovers that the hoarding behavior of institutions inflicts on other institutions. Relative to a model without hoarding or strategic reactions to intraday reserves imbalances, the liquidity-hoarding behavior of other banks collectively contributes to additional non-payments. An individual bank, by halting payments, can retain liquidity, leading to *endogenous* liquidity traps. These traps can, in turn, exacerbate the reduced flow of payments, as other banks, which may not have been significantly impacted by the first wave, also pass their trigger strategy and amplify the initial liquidity shortage. Importantly, from a counterparty's perspective, the primary institution that is impaired due to the cyber attack, and secondary institutions that hoard liquidity may be indistinguishable. As a result, even if the network distance between an institution and the primary target institution is large, the institution trigger strategy is aptly described as before — either due to concern about a cyber attack or prompted by liquidity concerns.

Of course, banks' endogenous reactions to intraday reserves imbalances, as well as their reaction to the failure to receive payments from counterparties, can be modeled in other ways. We believe this approach is parsimonious and captures the dimensions through which banks may respond to inform the scale of the cascade. One can also envision a scenario in which a malicious attacker publicizes the attack, and even banks that were not exposed to any missed payments begin to hoard liquidity, in which case the cascades would be even larger than those we model here.

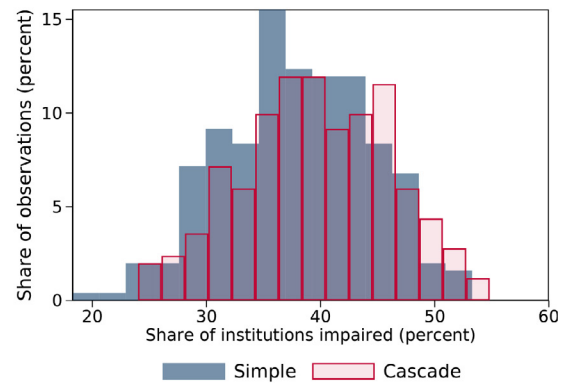
5.2. Impairment in cascades

When banks respond to intraday payment activity, banks, responding to abnormal liquidity shortages from

⁸ Note, this approach implies that for each institution, there is exactly one day in which it reaches $\eta_{\max}^i$. To avoid mechanical liquidity hoarding, we exclude the institution from reactions on the day that $\eta_{\max}^i$ is actually reached, though doing such does not significantly affect the results.



(a) Unweighted share.



(b) Weighted share.

Fig. 5. Empirical distribution of network impact for cascade scenario. The figure shows the distribution of the network impact for the baseline scenario (blue) and the cascade scenario (red), averaged across the top-5 institutions. The left panel shows the unweighted share of impaired institutions. The right panel shows the share of impaired institutions weighted by assets. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

deficits in payment inflows, may individually seek to improve their liquidity positions by forgoing payments. Although this reaction imposes network externalities, the overall impact on liquidity dislocations is unclear. Institutions may collectively resolve uncertainty about their end-of-day liquidity conditions by hoarding liquidity. Setting aside the large quantity of unsettled payment demand, if more institutions are able to avoid liquidity problems by hoarding liquidity, fewer banks may be impaired, as defined in Section 4.1, as a result of the cyber attack.

Our results suggest liquidity hoarding does not lower the share of impaired institutions. Fig. 5 shows the empirical distribution of the cascade scenario and compares it with the baseline scenario. In both, we compare averages over each of the days for an attack on each of the top-5. We find strategic liquidity hoarding, if anything, amplifies the network impact of the cyber event. This finding indicates that, taking into account strategic behavior of payment network participants, a cyber attack has even greater systemic consequences.

The cascade scenario provides us with a high-frequency counterfactual illustrating how liquidity hoarding evolves over the course of a given day (Fedwire payments for day t occur between 9:00pm EST on day $t-1$ and 6:30pm EST on day t). Fig. 6 illustrates results on intraday dynamics of liquidity-hoarding behavior from two cascade scenarios (attacks on top-5 banks A and B), averaged across days in 2018. Over the course of the day, more and more counterparties of the attacked institution pass their trigger threshold, and start to hoard liquidity. Eventually, their suspended payments also cause their recipients to start hoarding, and so on. The pace at which hoarding accelerates over the day is driven by a combination of the primary and secondary channels. First, on a normal day, payment activity intensifies toward the afternoon. As a result, the original attacked institution's failure to remit payments affects a broader set of institutions later in the day. As the day unfolds, the number of institutions that switch to liq-

uidity hoarding increases, even more institutions pass their intraday trigger threshold and revert to self-preservation as well.

5.3. Forgone payments in cascades

A stark difference, relative to the baseline scenario, is in the value of payments that are not made as a result of hoarding behavior. Relative to before, the systemic implications are driven not only by the compromised liquidity positions of financial institutions, but by the failure to facilitate vital financial services and operations necessary for financial markets and the broader economy. Under the cascade scenario, forgone transactions represent from 5% to 35% of total daily payment value, an amount equal to one to 11 times daily GDP.

Fig. 7 provides a distribution of the average forgone transactions by payment value associated with cascade scenarios on the top-5 institutions. The range of the distribution illustrates the variability of the vulnerability in calendar time as discussed in Section 4.5. As expected, a significant fraction of forgone value can attributed to the shocked institution, which in our baseline scenario is a top-5 institution by payment value. The net forgone volume represents forgone payments by value for hoarding institutions, which still reaches 0.75 times daily GDP on average.

Wholesale payments such as those in Fedwire can be broadly categorized as payments related to real economic activity, for example, firms paying their suppliers, and payments related to financial activity, for example, interbank loans. We attempt to identify these two categories using a business function code that institutions can apply to each of their Fedwire payments. We proxy for real-activity payments by using the code "Customer Transfer" and for financial-activity payments using the code "Bank Transfer."

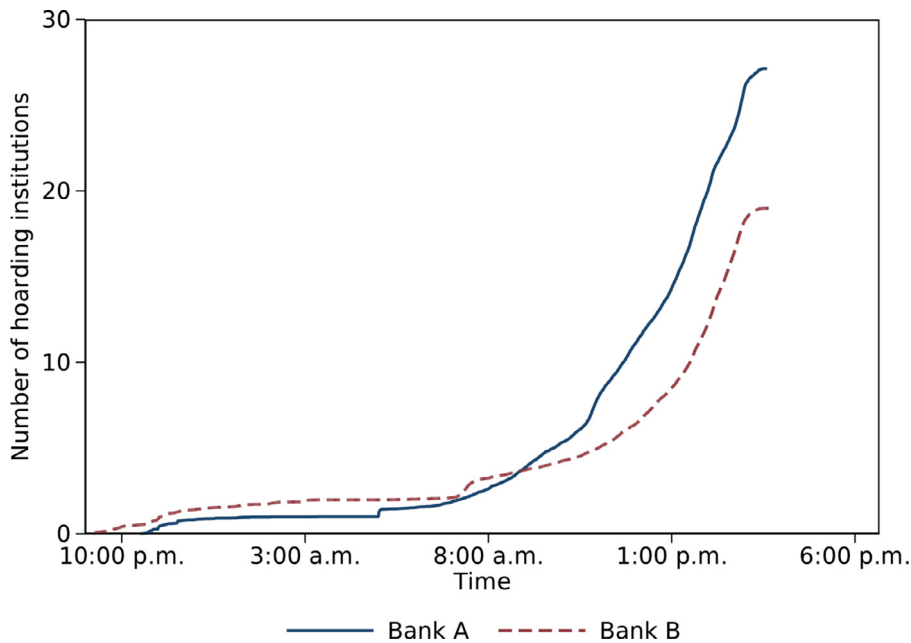


Fig. 6. Intraday dynamics of liquidity hoarding. The figure shows, over the course of the Fedwire day, the average number of institutions triggered to hoard liquidity for the cascade scenario of two top-5 institutions.

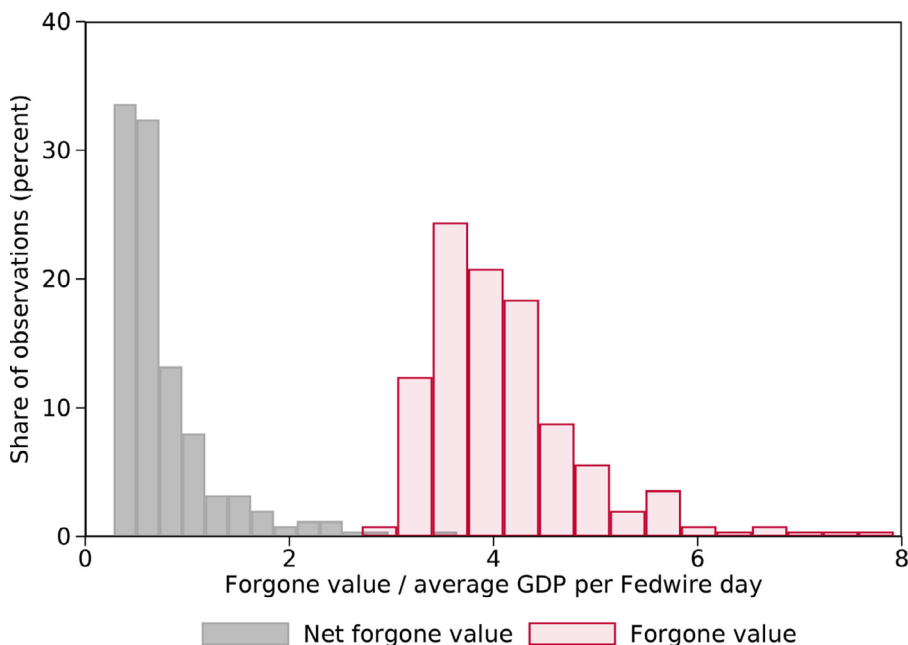


Fig. 7. Distribution of total value of forgone transactions. The figure shows the distribution of the value of transactions forgone as a consequence of banks' liquidity-hoarding behavior in the cascade scenario, averaged across the top-5 institutions. Payment value is scaled by daily GDP, where the number of days is given by Fedwire days. The net forgone value excludes payments by the shocked institution.

Fig. 8 shows the distribution of payments across real and financial activity in normal times (blue bars) and compares it with the distribution of forgone payments in the cascade scenario (red bars). We see, first, that the distribution of forgone payments varies dramatically more across days than the distribution of normal payments (whiskers). This observation is consistent with the high variance of

network impact across days and the implied tail risk for cyber events (Section 4.5). In addition, we see that transactions with other financial intermediaries ("Bank Transfers") represent the majority of forgone payments and are over-represented compared with the distribution of normal payments, both in terms of value (Fig. 8a) as well as in terms of volume (Fig. 8b). This feature may be viewed as attenu-

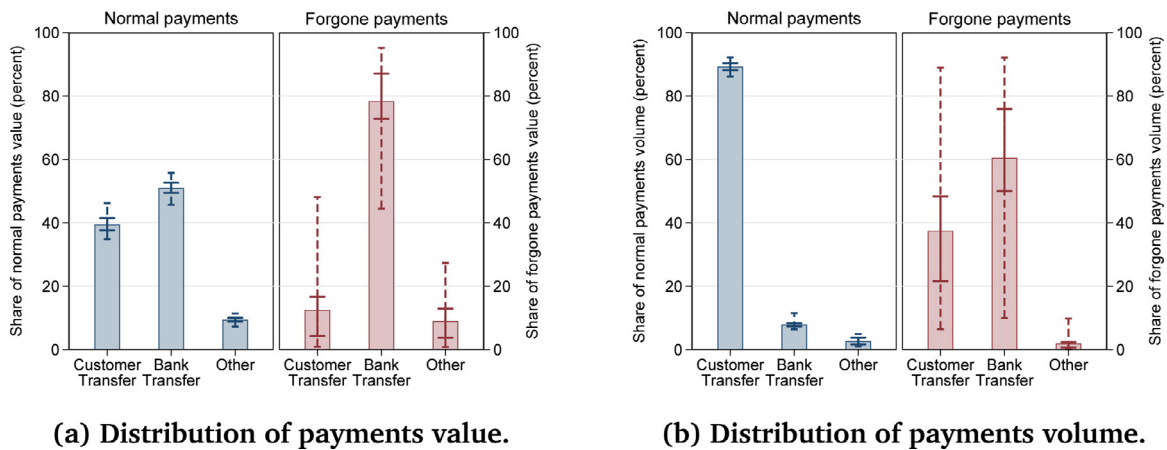


Fig. 8. Distribution of transfer types in normal payments and in the cascade scenario's forgone payments. The figure shows the distribution of Fedwire business function codes used among payments in normal times (blue), and the distribution used among forgone payments in the cascade scenario (red, averaged across the top-5 institutions). Fig. 8a shows the distribution in terms of payment value and Fig. 8b in terms of payment volume. Bars represent the average impact across days; solid whiskers represent the p25/p75 range, and dashed whiskers, the p1/p99 range across days. All statistics are "net" (excluding payments by the shocked institution). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ating the impact of the cyber event if we think such payments can more easily be handled via back-up systems, for example, because of high value but low volume or because of a high potential for netting. However, financial-activity payments could also be assigned a high multiplier in terms of real activity they support indirectly.

6. Correlated vulnerabilities

In this section, we explore scenarios in which multiple institutions are directly affected due to technological or other commonalities. As in the baseline scenario, we assume in the analysis in this section that banks do not respond strategically throughout the day.

6.1. Technological commonality

Existing research on financial networks examines transmissions between banks linked through interbank borrowing and other counterparty relationships. In the context of cyber, a salient concern is risk associated with technological commonality between financial institutions. As linkages between institutions in a network arise from financial motives, a financial network may not be structured to be resilient to shocks arising from technological similarities.

Third-party service providers present vulnerabilities especially when a provider is common to many institutions. However, third-party providers servicing many small institutions may offer improved cyber resiliency relative to what could be achieved by small institutions individually. For example, a third-party service provider may have the economies of scale or may be better at specializing in cyber security, thereby offering a safer option to smaller organizations. At the same time, with third-party service providers, a classical free-rider problem can arise because agents may individually have weaker incentives to monitor the common provider, which erodes governance and oversight (Shleifer and Vishny, 1997). Further, effectively moni-

toring the resilience of their third- (and fourth- and fifth-) party providers may be practically challenging for banks.

The Federal Reserve has highlighted the importance of managing risk from financial institutions' service providers, including activities beyond traditional core bank processing and IT services.⁹ Although the Federal Reserve supervises some key third-party service providers, the connections between institutions from shared service providers are not likely to be widely understood by the network.

We analyze a scenario in which several institutions share a service provider, making use of regulatory data on customer lists of significant service providers. Providers that serve the largest banks are effectively captured in the baseline scenario in Section 4. Some providers serve mostly smaller banks, which is captured by the reverse stress test scenario in Section 6.2. For the scenario in this section, we therefore select a provider with a mix of clients that provides IT services and software across industries, including the banking system. A vulnerability in one of its critical services that is exploited by an attacker could affect operations for exposed customers. We therefore assume that a successful cyber attack on the service provider disrupts all client institutions' payment operations as if they were directly attacked.

Fig. 9 shows the empirical distribution of the scenario and shows a shock to the set of large and medium banks can have an outsized impact on the payment network. Similar to the baseline scenario, the weighted share of impaired institutions is several times greater than the unweighted share. However, the overall magnitude is greater than the effect of an attack on a single top-5 institution with an average weighted impact of 60%. This finding points to the potential for a third-party service provider, by simultaneously impairing multiple institutions, to cause a

⁹ See, for example, SR-13-19 Guidance on Managing Outsourcing Risk (<https://www.federalreserve.gov/supervisionreg/srletters/sr1319.htm>).

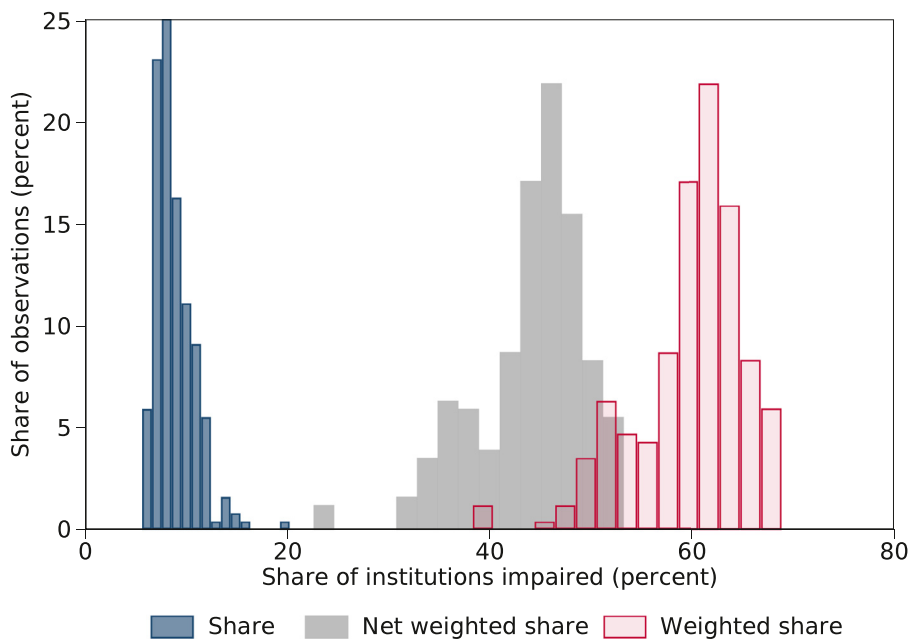


Fig. 9. Cyber vulnerability through a third-party service provider. The figure shows the distribution of the network impact for the scenario with a disruption originating from a third-party service provider. “Share” represents the unweighted percent of institutions that become impaired. “Weighted share” represents the percent of institutions that become impaired, weighted by asset size. “Net weighted share” refers to the percent of institutions that become impaired, net of the shocked institutions.

systemic event that is amplified through the payment network and propagates through the U.S. financial system.

6.2. Reverse scenario

A clear implication of the baseline scenario is that a successful cyber attack on any single one of the top-5 institutions has the potential to have significant spillovers to other banks, amplifying the attack through the financial system. However, these large institutions can take advantage of economies of scale in investment in cyber defenses and, as systemically important institutions, they are also subject to heightened supervisory standards and stringent capital and liquidity regulation. They also dedicate large amounts of resources to cyber resiliency, meaning that from the perspective of a malicious attacker, a cyber attack on the largest institutions would be costly. A malicious actor may therefore instead target institutions that may be relatively easier to successfully infiltrate. In particular, smaller institutions may have fewer resources to defend against a sophisticated attack. With this in mind, we run a *reverse scenario* — estimating what is the minimum set of smaller institutions necessary to impair a top-5 institution in the payment network.

With a large amount of time and computing power, it would be possible to estimate all the different permutations of institutions that would succeed in impairing one of the top-5 institutions, or resulting in the impairment of a given share of payments or banking assets. To implement this approach more parsimoniously, we group institutions by asset size, as a number of regulations and supervisory standards are implemented with cut-offs based on asset size. Implicitly, this analysis highlights that super-

visory and regulatory standards based on bank size may not take into account the network externalities of smaller banks with respect to a cyber attack.

Our reverse scenario is implemented as follows. Two sets of potentially shocked institutions, $U_N \subset U$, are considered: (i) “mid-sized” entities with asset size between \$10 billion and \$50 billion, denoted U_{50} ; (ii) “small-sized” entities with asset size below \$10 billion, denoted U_{10} . These cutoffs are chosen to identify subsets of banks under meaningfully different and progressively relaxed regulatory and supervisory conditions.¹⁰ In setting these cut-offs, we are motivated by Federal Reserve Supervisory guidance such as that laid out in SR 12-17,¹¹ which makes size-based distinctions in a supervisory framework that differs at the \$50 billion in assets cut-off. This framework specifically does not apply to community banking organizations, defined as banks under \$10 billion in size. Also, the Dodd-Frank Act only requires banks with more than \$10 billion in total assets to be subject to regulatory oversight by the CFPB and to be required to perform and report the results of annual firm-run stress tests. To focus on small banking institutions, we exclude any accounts associated with government-sponsored entities (including federal home loan banks), FMUs, and branches of FBOs.

For each set of potentially shocked institutions $U_N \in \{U_{10}, U_{50}\}$, for each top-5 institution j , and for each day t , we identify the smallest subset of U_N that if successfully attacked, would lead to the top-5 institution becoming impaired, according to the impairment definition in

¹⁰ Eisenbach et al. (2021) provides a detailed study on bank supervision based on size.

¹¹ <https://www.federalreserve.gov/supervisionreg/srletters/sr1217.htm>

Table 3

Summary statistics of reverse scenario.

The table reports the distribution for the reverse scenario of $|U_{Nt}^{\min}|$ for entities under \$10 billion in asset size, $N = 10$, and entities between \$10 billion and \$50 billion in asset size, $N = 50$. Panel (a) excludes branches of FBOs and panel (b) includes them. “Days with Impairment” indicates the number of days for which at least one set of banks in U_N existed for which a top-5 institution would become impaired.

Impairment	p1	p25	p50	p75	p99	Mean	SD	Days with Impairment
(a) Excluding branches of FBOs								
U_{10}	1	1	5	24	221	24	50	23 of 250
U_{50}	1	1	3	8	111	10	25	101 of 250
(b) Including branches of FBOs								
U_{10}	1	1	2	6	381	11	43	180 of 250
U_{50}	1	1	1	1	6	1	1	250 of 250

Section 4.1. Formally, we identify the set $U_{Njt}^{\min} \subset U_N$, which is given by

$$U_{Njt}^{\min} = \arg \min_{U' \subset U_N} |U'| \text{ s.t. } j \in B \text{ for } (U', R, B) \text{ on date } t. \quad (4)$$

We then identify for each day the minimum number of entities required to impair any one of the top-5 institutions, that is the set $U_{Nt}^{\min} = \arg \min_j \{|U_{Njt}^{\min}|\}_j$. We outline the results of the reverse scenario in Table 3, panel (a). The table outlines both the number of days in which a top-5 institution is impaired or not, along with summary statistics of the number of institutions in $U_{Nt}^{\min}$ for t in 2018.

Our results suggest a successful attack on small institutions could be amplified dramatically through its effect on top-5 institutions. On 23 of the 250 Fedwire days in 2018, a coordinated cyber attack on a set of small institutions, if judiciously selected, could result in at least one of the top-5 institutions' reserves falling below its liquidity threshold. Among those days on which a potential impairment arises, an average of 24 small institutions successfully attacked were sufficient to impair at least on top-5. Raising the size to mid-sized entities suggests even more amplification; we find a greater number of days with potential impairments, and a smaller number of successfully attacked institutions required to impair a top-5 institution.

The amplification from small banks can be attributed to two factors. First, though small in size, these institutions may collectively have large payments by value. Second, on certain days, end-of-day reserves of a top-5 institution may be close to its impairment threshold. On such a day, the top-5 institution's end-of-day reserve buffers may be insufficient to cover unexpected payment deficits even from small institutions. Combined, these factors suggest that vulnerability can be reduced both through strengthening the cyber resilience of small institutions and through robust liquidity buffers for the largest institutions.

To the extent that supervisory scrutiny can help ensure cyber resilience, we also explore the impact of a potential attack on branches and agencies of FBOs. These firms have access to Fedwire payments, but, much like small U.S. banks, are not subject to the heightened supervisory scrutiny at large domestic institutions. Furthermore, many FBO branches are “small” by assets, but exhibit outsized payment activity.

Table 3, panel (b) shows results of our reverse scenario when we allow the sets U_{10} and U_{50} to include FBO branches. After including FBO branches, we find a subset of banks results in an impairment of a top-5 institution for 72% (U_{10}) and 100% of days (U_{50}). For the majority of Fedwire days in 2018, two or fewer institutions successfully attacked are sufficient to impair at least one of the top-5 institutions. Strikingly, the reverse scenario also indicates that, on an average day in our sample, only one mid-sized institution in U_{50} is necessary to impair a top-5 institution. Several factors drive these results. First, FBO branches are responsible for a significantly larger value of payments relative to their asset size. Second, our results indicate greater payment flow between FBO branches and core institutions relative to a domestic institution of similar size. The large role that these institutions play in the payments network suggests ensuring their cyber resiliency in the context of the U.S. payment system is important.

7. Feedback and amplification to other financial markets

An integral feature of the financial system is the interdependence of multiple markets, often linked by institutions and operations. Because the wholesale payment system includes payments to other market infrastructure entities, we can trace the implications of cyber attacks beyond the payments system. Examples of such other market infrastructures are exchanges, central counterparties, and other payment systems, commonly referred to as FMUs. The top-5 participants in the wholesale payment system are also key players in the operation of FMUs. As such, the underlying links between critical financial networks can serve as a nexus through which cross-exposures arise. In this section, we consider the potential for feedback and amplification in a multi-network environment involving Fedwire and FMUs.

The impact of a cyber attack that disrupts payments in Fedwire can interact with FMUs through two channels. First, a cyber attack can indirectly affect an FMU if payment disruptions in Fedwire impair the FMU's liquidity and thereby propagate and trigger settlement failures and illiquidity in financial markets more broadly. This indirect effect is additive to the direct effect on the FMU of the impairment of a large participant. Second, a cyber attack can directly affect the operations of FMUs, which of-

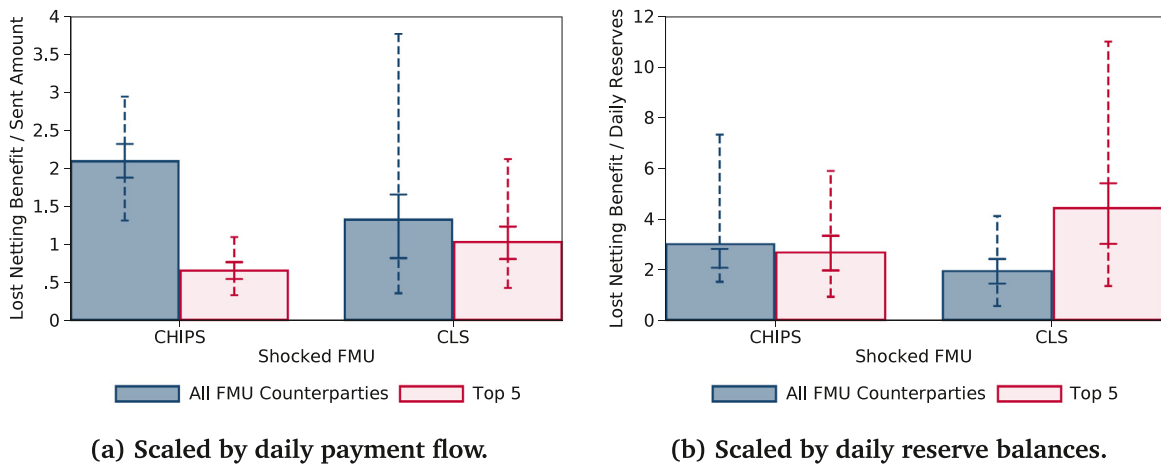


Fig. 10. Distribution of failed payments. This figure shows the distribution of the estimated value of failed payments on CHIPS and CLS, scaled by daily Fedwire payments made by each bank (panel a) and by daily reserve balances of each bank (panel b). Bars represent the average ratio; solid whiskers represent the p25/p75 range; dashed whiskers the p1/p99 range. The blue bar includes all member banks of the DFMU, and the red bar includes only top-5 banks. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

fer unsubstitutable services such as netting.¹² These services can lead to a sudden and severe scarcity in liquidity-intensive operations, namely, wholesale payment and settlement, creating an illiquidity spiral.

We begin with the first channel, estimating the disruption in payments sent to designated FMUs (DFMUs) within our main scenario (attack on a top-5 institutions).¹³ Each of the top-5 institutions accounts for a sizable fraction of total daily payments received by DFMUs, namely, 12% on average. A successful cyber attack on a large U.S. institution would thus also have a significant impact on the liquidity of the DFMU, which would also potentially imply an additional liquidity constraint on non-affected DFMU participants, as many DFMUs maintain credit line agreements to access liquidity, in many cases provided by DFMU participants.

We also analyze the second channel – a cyber attack that directly affects a DFMU – and how it might affect the financial system. We consider two key DFMUs: the Clearing House Interbank Payment System (CHIPS) and Continuous Linked Settlement (CLS). CHIPS is a wholesale payment system that offers multilateral netting benefits, and therefore enables banks to economize on liquidity relative to the real-time gross settlement of Fedwire. CLS provides settlement across different currencies' payment systems and is a key part of the infrastructure for global foreign exchange markets. As the members of CHIPS or CLS also have access to the Fedwire network, institutions can substitute payments meant to occur on these networks onto Fedwire. However, the key benefits FMUs offer are not substitutable. Specifically, institutions would no longer be able to realize the liquidity and capital savings of netting (CHIPS and

CLS) and counterparty risk protections (CLS).¹⁴ If banks are forced to settle activity without access to FMUs, the system as a whole could experience a severe shortage of liquidity with a corresponding impact on intermediation of market trades.

Our analysis should be interpreted as a lower bound on the potential for feedback and amplification through DFMUs and market infrastructure more broadly. We do not explicitly consider the possibility of margin increases after the cyber attack or the possibility of default funds being drawn on, both of which would generate additional liquidity stress. Further, we do not consider potential precautionary responses of other counterparties that have the potential to further increase any stress on a market infrastructure. We also do not quantify the impact of a CCP disruption on financial markets.

Although we are not able to directly observe the underlying transactions between institutions within the systems of each of these DFMUs, both CHIPS and CLS ultimately rely on Fedwire to settle participants' net payment obligations. We are therefore able to observe the flows between banks and DFMUs on Fedwire and combine our observations with information on gross DFMU activity from public data. The gross value of payments processed within CHIPS and CLS are large, each reaching about \$1.7 trillion daily.¹⁵ To shed light on the magnitude in the liquidity deficit resulting from an attack on an DFMU, we calculate the netting benefits of the DFMU by taking the ratio of the (gross) activity on the DFMU to the (net) flows to the DFMU on

¹² For example, Fleming and Keane (2021) estimate the impact of interdealer netting of Treasuries in the central clearing party (CCP) to be almost \$300 billion, on average. Therefore, in a cyber attack on that CCP, participants would forego those netting benefits.

¹³ For a list of DFMUs, see https://www.federalreserve.gov/paymentsystems/designated_fm_u_about.htm.

¹⁴ In the case of CLS, an added complication is the netting of transactions involving multiple currencies, which requires coordination across multiple central bank wholesale payment systems.

¹⁵ For CHIPS, we use the average daily gross value of payment in 2018, reported on the CHIPS website. CLS publicly reports daily gross value at the monthly frequency. Hence, the analysis on CLS is based on the average daily gross value of payment in each month of 2018.

Fedwire.¹⁶ Then, we can approximate the value of additional payments each bank would have to divert to Fedwire if the DFMU were not available. On each day for each bank, we multiply the daily aggregate gross-to-net ratio by the each individual bank's daily payments to the DFMU.

Fig. 10 shows that for both CHIPS and for CLS, the additional payments value that would have to be executed on Fedwire would be massive, adding to normal Fedwire payments value by a factor of 2.11 to 1.34 on average, respectively. For CHIPS the ratio varies between 1.25 and 3.65, whereas for CLS, the variation is considerably larger, between 0.16 and 4.23.

In addition, we quantify the liquidity impact of losing the inoperable DFMU's netting benefits, by scaling the excess payments by the actual end-of-day reserves balances of each institution, shown in Fig. 10b. The reserves balance of each bank represents the bank's liquidity available to process payments. Scaling by end-of-day reserves balances enables us to gauge how constrained individual institutions may become in processing a sudden jump in payments. The excess payments amount to three and two times banks' daily reserve balances, on average, for CHIPS and CLS, respectively. This finding indicates a cyber attack, which results in an impairment of DFMU operations, whether direct or indirect, could lead to a sudden scarcity in system liquidity.

8. Conclusion

Our analysis demonstrates how cyber attacks on a single large bank, a group of smaller banks, or a common service provider can be transmitted through the payments system. A cyber attack on any of the most active U.S. banks that impairs any of those banks' abilities to send payments would likely be amplified to affect the liquidity of many other banks in the system. The extent of the amplification would be even greater if banks respond strategically, which they are likely to do if uncertainty exists regarding the attack. The impact on geographies with concentrated banks may be even larger. We also identify other ways that the system may become impaired that highlight the importance of all banks in the network, not just the largest banks. First, if a number of small or midsize banks are connected through a shared vulnerability, such as a significant service provider, a shock could be transmitted throughout the network. Similarly, banks with a relatively small amount of assets but large payment flows also have the potential to impair the system.

¹⁶ Note payment flow to the DFMU on Fedwire is a proxy for net payment activity on the DFMU. It represents the liquidity the DFMU requires participants to hold, which, in turn, is determined primarily by the participant's payment activity on the DFMU.

Although the shock we assume is extreme in some ways — a complete inability to send payments — it is conservative in others. Our extensions suggest potential spillovers outside the payment network, such as to short-term creditors that provide liquidity to impaired banks. Allowing creditors and customers to run without the ability to realize additional liquidity through the sale of liquid assets would surely make our results even larger. If a cyber attack were to compromise the integrity of banks' systems, the reconciliation and recuperation process would be an unprecedented task. A cyber attack could have severe implications on the stability of the broader financial system vis-vis spillovers to investors, creditors, and other financial market participants. One shortcoming of the analysis is that although we see payment flows among banks, we do not know the purpose of the flows. Therefore, we cannot measure how these flows will affect customers and borrowers and the real economy. In future work, we hope to explore how borrowers and other counterparties would respond.

Appendix A. Additional tables and figures

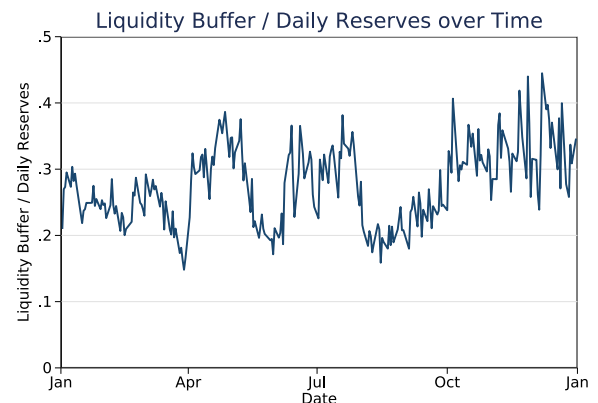


Fig. A.1. Liquidity buffer relative to daily reserve balances for top-5 banks. This plot shows how the liquidity buffer in the impairment threshold, the two standard deviations of banks' end-of-day reserve balances, varies over time, relative to the daily end-of-day reserve balances for the top-5 banks.

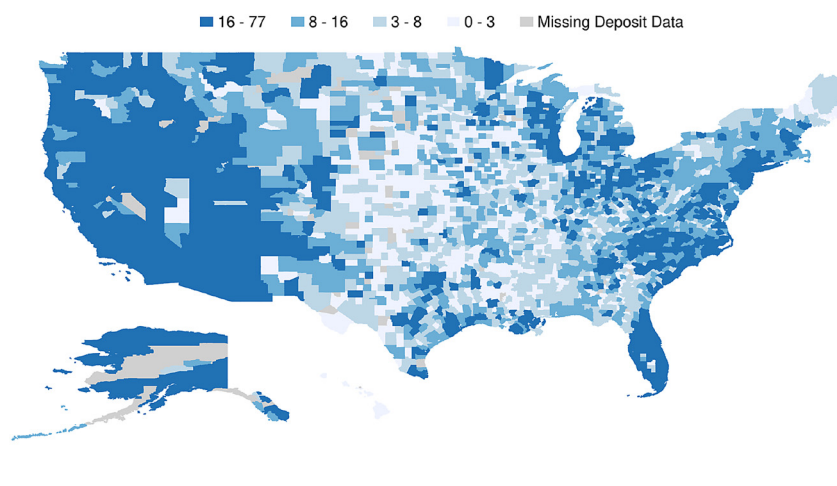


Fig. A.2. Geographic impact at the county level. The figure shows the average impact of the baseline scenario, where impact is measured by the deposit share (percent) of impaired institutions within each state. Impact excludes the shocked institution.

Table A1

Distance from impairment threshold. This table summarizes statistics on banks' average z-score for reserves over 2018, weighted by asset size. The z-score measures banks' distance, in standard deviations, from their 30-day rolling means in the sample period.

p1	p5	p25	p50	p75	p95	p99	Avg.	St. dev.
−.4996	−.4565	−.1570	−.0342	.0588	.2422	.6300	.4852	63.5251

Table A2

Summary statistics for regressions. The table shows summary statistics of the variables used in the regressions in Table 2.

	Avg.	St. dev.	p1	p5	p95	p99
Fraction of days impaired	0.074	0.164	0.000	0.000	0.350	1.000
Log assets	13.105	1.456	10.662	11.256	15.894	18.455
Liquid asset share	0.102	0.103	0.014	0.020	0.281	0.576
Unstable funding share	0.316	0.159	0.037	0.094	0.566	0.682
Capital ratio	0.110	0.043	0.065	0.081	0.157	0.220

Appendix B. Classifying impaired institutions

As a robustness exercise, we examine a variation that excludes from the set of impaired institutions any banks

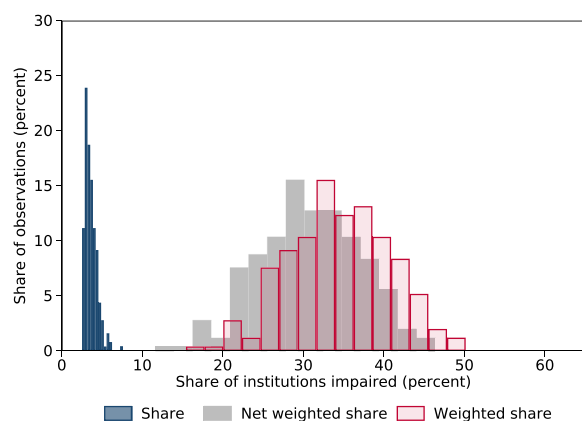


Fig. B.13. Average network impact for alternative impairment classification. The figure shows the distribution of the network impact from the baseline scenario where we exclude from the set of impaired institutions any banks that are impaired but do not receive any payments from the shocked institution on that day.

that are impaired but do not receive any payments from the shocked institution on that day. Fig. B.13 shows results for the average network impact, after excluding the subset of banks that satisfy the above criteria. The weighted share of impaired institutions remain large, whereas we see a noticeable decrease in the unweighted share of impaired banks. These results indicate that, consistent with the main analysis, that network impact is primarily driven by large, core institutions.

Appendix C. Impact of multi-day cyber attack

This appendix extends our analysis to evaluate the multi-day impact of a cyber attack. We approach the multi-day scenario by extending our baseline scenario from one to up to five days, under the assumption that banks, excluding the shocked institution, continue to send payments as usual. Banks' payment behavior would likely change following a successful cyber attack (e.g., as in the cascade scenario in Section 5). However, our approach allows us to analyze how the severity of liquidity dislocation may grow with each incremental day of operational disruption, and the nature of emergency liquidity support that may be required in the aftermath of a cyber attack. It also reflects our preference to keep the scenario as simple as possible to allow for clear interpretation.

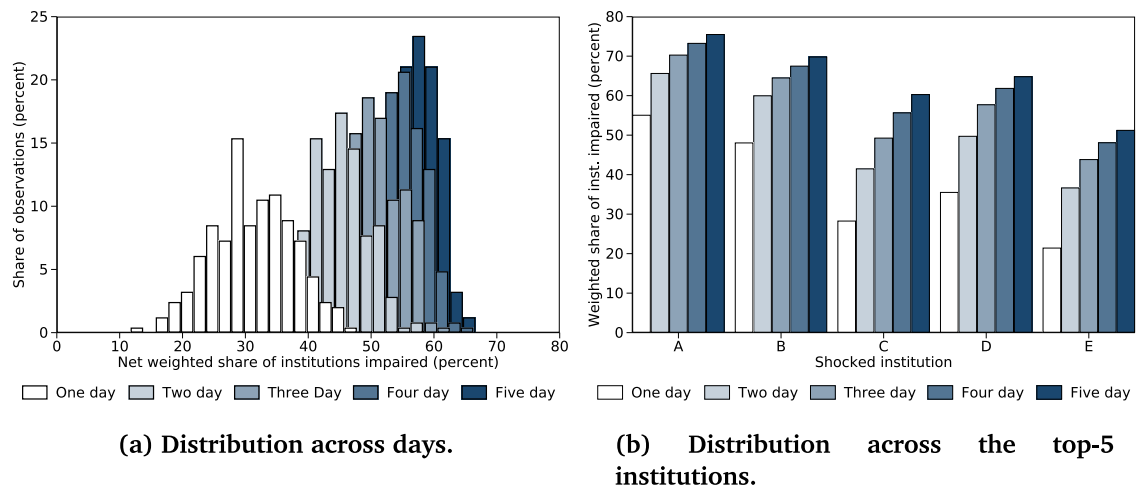


Fig. C.14. Multi-day impact of a shock to the top-5 institutions. The figure shows the distribution of net weighted share of impaired institutions for an attack that lasts from 1 to 5 days for each of the top-5 institutions. Panel (a) shows the distribution across days, averaged across the top-5 institutions; panel (b) shows the distribution across the top-5 institutions, averaged across days.

We extrapolate our baseline scenario to consider the cumulative set of impaired institutions as the shocked institution remains inoperable for up to five days. Following the shock, with each incremental day, we identify the additional set of institutions that become impaired. As a result, the x -day net weighted share is equal to the share of institutions, weighted by assets, that become impaired as a result of payment deficits arising from day one to day x . We adapt the single-day impairment definition for the multi-day scenario: an institution is considered impaired in the scenario if, within the x days, the end-of-day counterfactual reserve balance falls below the impairment threshold of that given day.¹⁷

Figure C.14 summarizes the average network impact across the top-5 scenarios, as the disruption lasts from one to five days. Two observations regarding the changes in the distribution are notable. First, the distribution shifts to the right as the disruption to the shocked institution persists for longer periods of time, with the average net weighted share of impaired institutions increasing from more than 30% within the first day to almost 60% by the fifth day.

Second, the distribution becomes *more* concentrated as the disruption is prolonged. This feature is evident when breaking down the network impact by scenarios of each of the top-5, shown in Fig. C.14b. Once a shock persists for five days, the network impact of an attack on any of the top-5 impairs over 50% of the banking sector by assets, and the differences across the top-5 are considerably smaller than on day one.

This finding is partly due to the fact that prolonged disruptions enable the shock to propagate further throughout the network. From the first to the second day, the average size of impaired institutions almost doubles from \$44 billion to \$85 billion in assets as the shock is transmitted to an increasing number of core institutions with high payment value. Over the next days, the average size of in-

crementally impaired institutions decreases, reaching \$28 billion on average by the fifth day. Overall, this observation suggests liquidity issues spread first through the core and periphery institutions dependent on the shocked institution, and over time, reach other periphery banks within the network.

An alternative method of evaluating the network impact in a multi-day scenario is to evaluate the severity of the liquidity dislocations. For a multi-day scenario, institutions' reserve holdings may drop below material supervisory thresholds. In such an event, extraordinary actions would be required by regulators and the discount window to ensure banks operate as usual.

Figure C.15 shows the share of institutions whose balance falls below zero becomes significant over time, both in terms of the raw and the size-weighted share. By the fifth day, the share and weighted share of institutions with negative balance both reach 37%, on average, and up to a maximum of 47% and 45%, respectively.

Because our analysis shows nearly half of institutions would run out of reserves by day five, it is useful to quantify the liquidity support that would be required to restore reserve balances to the impairment threshold of the baseline analysis. For each multi-day scenario, we calculate the liquidity shortfall in the system, defined by the gap between institutions' impairment thresholds and their counterfactual reserve balance, aggregated across all impaired institutions. We find that following a cyber attack, as shown in Fig. C.16, the average liquidity shortfall grows from \$122 billion on the first day to an average of \$1 trillion by the fifth day. In the 2018 sample period, this growth would have implied an expansion of the Federal Reserve balance sheet of over 20%, highlighting the magnitude of liquidity injections required just to support regular payment activity. This estimate also approximates the potential disruption that would arise if the books and records of an institution were impaired, because liquidity provision would not resolve issues regarding payment allocations at the bank-customer level.

¹⁷ Varying the multi-day impairment definition does not yield meaningful differences in the results.

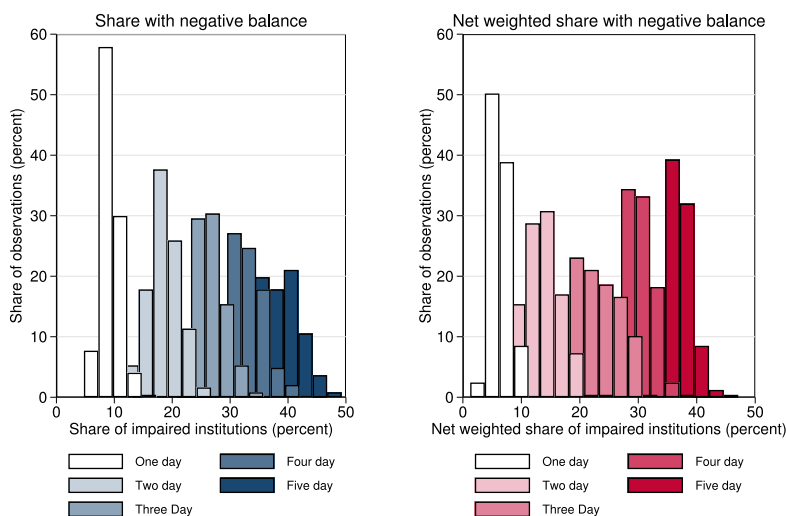


Fig. C.15. Share of institutions with negative counterfactual reserves balance. The figure shows the distribution of the share of institutions whose counterfactual end-of-day reserves balance drops below zero as a result of an attack on a top-5 institution that lasts 1 to 5 days. The left panel shows the average share across the top-5 scenarios, including the shocked institution. The right panel shows the results excluding the shocked institution.

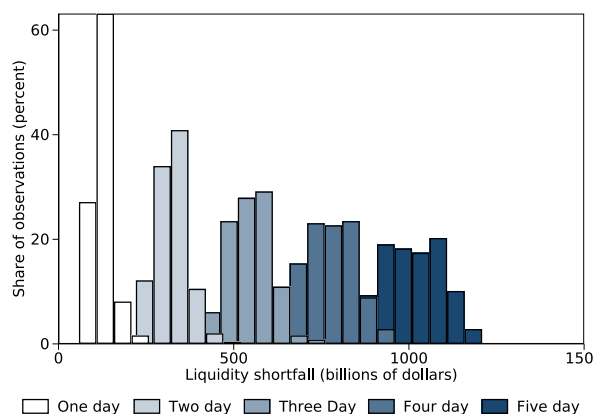


Fig. C.16. System liquidity shortfall. The figure shows the distribution of aggregate liquidity shortfall resulting from an attack that lasts from 1 to 5 days, averaged across the top-5 scenarios.

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